A FIXED-MASK PERMUTATION MODEL FOR THE MÖBIUS SUMMATORY PATH
EXACT ORBIT CALCULUS AND AN ORBIT-NORMALIZED RH CRITERION

ABSTRACT. For a fixed cutoff N, the nonzero values of the Möbius function form a finite sign word on the squarefree support. We study the uniform orbit obtained by permuting those signs while preserving the support and the two sign counts. This is an exchangeable hypergeometric bridge with the same endpoint $M(N)$ as the observed word. For the normalized logarithmic quadratic path mass

$$\mathcal{Q}_{N_0,N}(w) = \frac{1}{\log(N/N_0)} \sum_{N_0<n\le N} \frac{R_w(n)^2}{n^2}, \qquad R_w(n) = \sum_{k\le n} w(k), \tag{0.1}$$

we derive exact finite-class formulae for the mean, the covariance kernel of squared partial sums, and the variance. The covariance is semiseparable, with tridiagonal inverse on every nonsingular positive block. We also give an exact bilinear decomposition, an opposite-sign transposition calculus, a closed form for the global minimum and all minimizers, and the M-convex local–global exchange property. The full cumulative quadratic profile determines the absolute path exactly; its fiber through the Möbius path is the excursion-reflection orbit. On this fiber, the divisor-convolution defect becomes an explicit positive-semidefinite Gram form in the excursion-orientation bits. An exact convolution identity then quantifies how divisor residuals propagate to deviations from the Mertens path.

The finite model also supplies an analytic normalization. For the positive path functional

$$\mathcal{B}_N(w) = \frac{R_w(N)^2}{N} + \sum_{n<N} \frac{R_w(n)^2}{n(n+1)}, \tag{0.2}$$

the exact expected centered-bridge scale in the fixed-mask, fixed-count orbit is asymptotic to $\zeta(2)^{-1}\log N$. Consequently, RH is equivalent to subpolynomial growth of $\mathcal{B}_N(\mu)$ relative to that orbit scale. This is an orbit-normalized reformulation of the classical Mertens criterion, not a proof of RH, but it locates the RH boundary intrinsically inside the permutation model. Separately, for proportional windows $N_0 = \lfloor \alpha N \rfloor$, the quadratic path mass converges rigorously to a weighted Brownian-bridge quadratic functional whose Karhunen–Loève spectrum is given by an explicit Robin eigenvalue equation.

For the identified Möbius data file through $N = 455{,}052{,}511$, the reported full-cutoff values are

$$\mathcal{Q}_{1000,N}(\mu) = 0.029334, \qquad \mathbb{E}_\pi \mathcal{Q}_{1000,N}(\pi) = 0.56628, \qquad \rho_Q(N) = 0.0518. \tag{0.3}$$

The inequality between the first two quantities is a finite, data-dependent computer-assisted statement with a declared forward-error audit; it is not an asymptotic theorem. The approximate five-percent scale is largely explained, under standard additional hypotheses, by the classical spectral second moment of the Mertens function. The analytic contribution is therefore not a new value for that constant, but an exact conditional orbit calculus, an absolute-path/orientation decomposition, and a class-normalized expression of the RH growth threshold. Reported excursion, conditional-choice, and blind spectral experiments are included with their observed outcomes and inferential limitations. No result in the paper proves the Riemann Hypothesis or the weak Mertens conjecture.

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1. Purpose, claims, and novelty boundary

Let μ denote the Möbius function and

$$M(x) = \sum_{n \le x} \mu(n), \qquad Q(x) = \sum_{n \le x} \mu(n)^2. \tag{1.1}$$

The model asks a finite conditional question: after retaining the exact squarefree support and the exact numbers of positive and negative values, how does the order of the Möbius signs compare with all other admissible orders? It does not replace multiplicative number theory by a random-sign hypothesis. It supplies a null class in which the information removed and the information retained are explicit.

1.1. **What is established.** The mathematical core is not a collection of checks at the reported cutoff. It consists of theorems valid for every finite fixed-support sign orbit, together with an asymptotic consequence of the entire sequence of Möbius orbits:

(i) the exact mean of (0.1);
(ii) a factorization of $\mathrm{Cov}(T_j^2, T_k^2)$ and an $O(m)$ formula for the exact variance;
(iii) an exact decomposition into a class-invariant diagonal term and an order-dependent bilinear term;
(iv) an exact formula for the change under one opposite-sign transposition, the global minimum and all minimizers, and M-convex local–global optimality;
(v) a semiseparable covariance representation with tridiagonal inverse on nonsingular positive blocks;
(vi) the reflection orbit of completed excursions and an exact formula for the mean number of zero returns;
(vii) the diffusive, Brownian-bridge critical scale of the null class.
(viii) an exact identification of the full quadratic-profile fiber with absolute-path geometry and excursion orientations;
(ix) an exact Gram representation of the divisor-convolution cost of those orientations;
(x) a rigorous proportional-window Gaussian quadratic limit with explicit spectrum;
(xi) an exact divisor-residual convolution identity and its weighted stability bound;
(xii) an orbit-normalized criterion in which RH is equivalent to subpolynomial growth relative to the exact centered-bridge scale.

The numerical component places the observed finite Möbius word in this class at fourteen cutoffs. The strongest data statement is the finite ratio in (0.3), not a limit as $N \to \infty$.

1.2. **What is new, and what is classical.** The finite-population bridge, its concentration inequalities, and its Brownian-bridge limit belong to standard probability theory [13, 24, 23, 1]. The weak Mertens conjecture, the conditional spectral second moment, and negative moments of ζ' are classical analytic number theory [26, 21, 9]. No priority is claimed for a general finite-population identity in isolation.

General quadratic forms under permutation and quadratic-assignment methods already possess an exact-moment literature [12, 14, 28]. The contribution asserted here is therefore narrower: the particular fixed-mask, fixed-count orbit for the Möbius word; the factorization and one-pass variance formula for its nested-max partial-sum kernel; the local ordering calculus; the exact absolute-path fiber and its arithmetic orientation Gram matrix; and the finite placement of the observed word against that exactly specified class. No claim is made that exact moments of a generic permuted quadratic form are new.

Discrete convex analysis and one-pair matrix algebra are likewise established theories. The model-specific statements are the closed-form lower envelope for this endpoint-constrained prefix functional, its application of laminar M-convexity to the sign orbit, the semiseparable specialization of the exact squared-prefix covariance, and the explicit Robin spectrum obtained from the logarithmically weighted proportional-window limit. The divisor-residual identity is elementary Dirichlet-convolution algebra; its role is to replace an unspecified selection remainder by a concrete stability operator, not to claim a new inversion formula.

The orbit-normalized RH equivalence proved in Section 6 has a different novelty status. Its denominator is new to this model and gives an intrinsic probabilistic scale, but the numerator and the final equivalence use the classical positive Mertens criterion. It is therefore claimed as a new formulation and interface between the permutation orbit and analytic number theory, not as a new zero-free-region estimate. This distinction is essential: the exact orbit theorems add finite algebra and algorithms, whereas the RH equivalence adds a coordinate system for a known unresolved condition.

Accordingly, the infinite-N content developed here concerns the permutation family: the asymptotic orbit scale, the aggregate logarithmic occupation measure, and the proportional-window Gaussian quadratic limit with explicit spectrum. These are presented as model-specific specializations, not as a claim that Brownian quadratic functionals or discrete convex analysis are new. No new unconditional asymptotic estimate for the particular function $M(x)$ is claimed.

Remark 1.1 (The meaning of the five-percent observation). The finite value 0.0518 is not merely a check that a known conjecture holds on a finite interval. It compares two different objects: the observed Möbius path and the exact mean of its own fixed-mask, fixed-endpoint orbit. That particular comparison appears to be specific to the present model. However, the numerator's scale near 0.029 is conditionally explained by the classical spectral constant in Section 6. Consequently, the class-relative normalization is the distinctive part; the small absolute second moment is not presented as an independent discovery.

Proposition 1.2 (Quadratic information selects an absolute path)**.** *A scalar quadratic functional cannot select the Möbius word uniquely. More precisely, the complete cumulative quadratic profile determines* $\{|R_w(n)| : n \le N\}$*, but no statistic depending only on the squares* $R_w(n)^2$ *can recover the independent orientations of completed excursions. Thus quadratic selection naturally acts on absolute-path equivalence classes, not on individual sign words.*

Section 4.2 proves this statement and identifies the reflection orbit as the exact fiber through the observed path. Section 8 then separates two questions: why the absolute path has small quadratic occupation, and what arithmetic information selects its orientation. This converts nonuniqueness from a defect of the statistic into a structural decomposition of the selection problem.

1.3. **Statistical status.** The permutation ladder, excursion samples, conditional-choice analyses, and blind spectral fit are reported computations. Exact zero-hit bounds use the one-sided Clopper–Pearson construction [3]. The conditional-choice p-values and the blind frequency matches remain exploratory until their raw outputs and complete selection pipelines are rerun independently. Appendix B specifies the required reproducibility package without executing it.

2. The fixed-mask permutation class

2.1. **Support, signs, and class hierarchy.** Fix $N \geq 1$ and write

$$A_N = \{n \leq N : \mu(n) \neq 0\} = \{a_1 < a_2 < \cdots < a_m\}, \qquad m = Q(N). \tag{2.1}$$

Let p and q be the numbers of $+1$ and -1 values on A_N. Then

$$p + q = m, \qquad p - q = S = M(N), \qquad p = \frac{m+S}{2}, \quad q = \frac{m-S}{2}. \tag{2.2}$$

Definition 2.1 (Fixed-mask and fixed-count classes)**.** Define

$$\Omega_N^{(0)} = \{w \in \{-1,0,1\}^N : \operatorname{supp}(w) = A_N\}, \tag{2.3}$$

$$\Omega_N^{(1)} = \left\{w \in \Omega_N^{(0)} : \#\{n : w(n) = +1\} = p\right\}. \tag{2.4}$$

$$\Omega_N^{(2)} = \left\{w \in \Omega_N^{(1)} : \left|\sum_{k \leq n} w(k)\right| = |M(n)| \text{ for every } n \leq N\right\}. \tag{2.5}$$

Equivalently, $\Omega_N^{(1)}$ is the orbit of $(\mu(1), \ldots, \mu(N))$ under the symmetric group acting on the values at the ordered support positions.

The two defining constraints are the zero mask and the sign multiset. The endpoint follows from them; it is not a third constraint. The first two cardinalities are

$$|\Omega_N^{(0)}| = 2^m, \qquad |\Omega_N^{(1)}| = \binom{m}{p}. \tag{2.6}$$

The entire function $Q(n)$, $n \leq N$, and the quantities p, q, S are class invariants. The intermediate summatory path is not.

For $w \in \Omega_N^{(1)}$, let

$$\xi_j = w(a_j), \qquad T_j = \sum_{i=1}^{j} \xi_i, \qquad T_0 = 0. \tag{2.7}$$

Then

$$R_w(n) = T_{Q(n)}, \qquad R_w(N) = T_m = S. \tag{2.8}$$

If

$$K_N = \#\{1 \leq j \leq m : T_j(\mu) = 0\}, \tag{2.9}$$

then Proposition 4.1 proves

$$\Omega_N^{(0)} \supseteq \Omega_N^{(1)} \supseteq \Omega_N^{(2)}, \qquad 2^m \geq \binom{m}{p} \geq 2^{K_N}, \qquad |\Omega_N^{(2)}| = 2^{K_N}. \tag{2.10}$$

Both inequalities are strict for the reported data at $N = 5 \cdot 10^6$, where the base-ten logarithms of the middle and final cardinalities are $915017.333\ldots$ and $1810.996\ldots$, respectively. At each level, the uniformization discards the divisor relations except for the invariants explicitly frozen by that level. The uniform measure on $\Omega_N^{(1)}$ is sampling without replacement from a population containing p values $+1$ and q values -1. Hence $(T_j)_{0 \leq j \leq m}$ is an exchangeable hypergeometric bridge; the finite-population moment framework is standard [4].

2.2. **Exact exchangeable moments.** Assume first that $m \geq 4$. For distinct indices define

$$\mu_1 = \mathbb{E}\xi_a = \frac{S}{m}, \tag{2.11}$$

$$\mu_2 = \mathbb{E}(\xi_a\xi_b) = \frac{S^2 - m}{m(m-1)}, \tag{2.12}$$

$$\mu_4 = \mathbb{E}(\xi_a\xi_b\xi_c\xi_d) = \frac{S^4 - 6mS^2 + 8S^2 + 3m^2 - 6m}{m(m-1)(m-2)(m-3)}. \tag{2.13}$$

The identities follow by expanding powers of $\sum_{j=1}^{m} \xi_j = S$, using $\xi_j^2 = 1$, and exploiting exchangeability. In particular,

$$\mathbb{E}T_j^2 = j + j(j-1)\mu_2. \tag{2.14}$$

For $m < 2$, the second mixed moment is unnecessary and the class is enumerated directly. For $2 \leq m < 4$, formula (2.12) remains available but fourth-order quantities are evaluated by direct finite enumeration; no undefined quotient is used.

Theorem 2.2 (Factored covariance of squared partial sums)**.** *For $m \geq 4$ and $1 \leq j \leq k \leq m$,*

$$\operatorname{Cov}(T_j^2, T_k^2) = \phi(j)\psi(k), \qquad \phi(j) = j(j-1), \tag{2.15}$$

where

$$\psi(k) = 2 + 4\mu_2(k-2) + \mu_4(k-2)(k-3) - \mu_2^2 k(k-1). \tag{2.16}$$

In particular, $\psi(m) = 0$, as required by the deterministic endpoint $T_m = S$.

Proof. Use

$$T_j^2 = j + 2\sum_{1 \leq a < b \leq j} \xi_a\xi_b. \tag{2.17}$$

In the product $T_j^2 T_k^2$, an unordered pair from the first prefix and an unordered pair from the second are identical in $\binom{j}{2}$ cases, share exactly one index in $2(k-2)\binom{j}{2}$ cases, and are disjoint in $\binom{j}{2}\binom{k-2}{2}$ cases. Exchangeability therefore gives

$$\begin{aligned}\mathbb{E}(T_j^2 T_k^2) = {} & jk + jk(j+k-2)\mu_2 + 2j(j-1) \\ & + 4j(j-1)(k-2)\mu_2 + j(j-1)(k-2)(k-3)\mu_4.\end{aligned} \tag{2.18}$$

On the other hand,

$$\mathbb{E}T_j^2\mathbb{E}T_k^2 = jk + jk(j+k-2)\mu_2 + jk(j-1)(k-1)\mu_2^2. \tag{2.19}$$

Subtracting (2.19) from (2.18) gives (2.15)–(2.16). The independent finite enumeration specified in Appendix B audits the resulting identity but is not used in the proof. □

2.3. **The critical square-root scale.** Put

$$U_j = T_j - \frac{jS}{m}, \qquad \sigma_m^2 = 1 - \frac{S^2}{m^2}. \tag{2.20}$$

For each fixed j, Hoeffding's comparison theorem for sampling without replacement gives

$$\mathbb{P}\left(|U_j| \geq \lambda\right) \leq 2\exp\left(-\frac{\lambda^2}{2j}\right). \tag{2.21}$$

Applying a union bound, rather than claiming a sharp maximal inequality, yields

$$\mathbb{P}\left(\max_{1 \leq j \leq m} |U_j| > \lambda\right) \leq 2m\exp\left(-\frac{\lambda^2}{2m}\right). \tag{2.22}$$

Thus, for every fixed $\varepsilon > 0$,

$$\mathbb{P}\left(\max_{j\le m}|U_j| > m^{1/2+\varepsilon}\right) \le 2m\exp\left(-\frac{m^{2\varepsilon}}{2}\right). \tag{2.23}$$

Lemma 2.3 (Uniform L^2 bound for the bridge maximum)**.** *For every nondegenerate fixed-count sign class with $m \ge 3$,*

$$\mathbb{E}_\pi\left[\left\{\frac{\max_{0\le j\le m}|U_j|}{\sigma_m\sqrt{m}}\right\}^2\right] \le 32. \tag{2.24}$$

Consequently the normalized maxima are uniformly integrable along every sequence of such classes.

Proof. Let $X_i = \xi_i - S/m$, let $\mathcal{F}_j$ be the first j revealed signs, and put $h = \lfloor m/2\rfloor$. Since the unrevealed centered population has sum $-U_j$,

$$\mathbb{E}(X_{j+1} \mid \mathcal{F}_j) = -\frac{U_j}{m-j}. \tag{2.25}$$

It follows that

$$Y_j = \frac{U_j}{m-j}, \qquad 0 \le j < m, \tag{2.26}$$

is a martingale. Doob's L^2 inequality and

$$\mathbb{E}U_h^2 = \sigma_m^2\frac{h(m-h)}{m-1} \tag{2.27}$$

give

$$\mathbb{E}\max_{0\le j\le h} U_j^2 \le m^2\mathbb{E}\max_{0\le j\le h} Y_j^2 \le 4m^2\mathbb{E}Y_h^2 \le 8\sigma_m^2 m. \tag{2.28}$$

Apply the same argument to the reversed centered sequence. Since its partial sums satisfy $\widetilde{U}_r = -U_{m-r}$, the second half is bounded by

$$\mathbb{E}\max_{h\le j\le m} U_j^2 \le 16\sigma_m^2 m. \tag{2.29}$$

The square of the full maximum is at most the sum of the two half-maxima. This gives the stated, slightly relaxed constant 32; the finitely many small boundary cases are checked directly. □

Proposition 2.4 (Finite-population bridge limit)**.** *Suppose $m \to \infty$ and $|S|/m \to s_0 < 1$. Put*

$$\mathbb{T}_m(t) = \frac{U_{\lfloor mt\rfloor} + (mt - \lfloor mt\rfloor)\{U_{\lfloor mt\rfloor+1} - U_{\lfloor mt\rfloor}\}}{\sigma_m\sqrt{m}}, \qquad 0 \le t \le 1, \tag{2.30}$$

with the fractional term interpreted as zero at $t = 1$. Then $\mathbb{T}_m$ converges weakly to a standard Brownian bridge B°. Consequently,

$$\frac{1}{\sigma_m\sqrt{m}}\max_{j\le m}|U_j| \Longrightarrow \sup_{0\le t\le 1}|B^\circ(t)|. \tag{2.31}$$

$$\mathbb{E}\max_{j\le m}|U_j| \sim \sigma_m\sqrt{\frac{\pi}{2}}\log 2\,\sqrt{m}. \tag{2.32}$$

Proof. The weak convergence is the finite-population functional central limit theorem; see [23]. The continuous-mapping theorem gives (2.31). Lemma 2.3 supplies uniform integrability, so convergence of the first moments follows. The limit distribution and its first moment use

$$\mathbb{P}\left(\sup_t|B^\circ(t)| \le x\right) = \sum_{r\in\mathbb{Z}}(-1)^r e^{-2r^2x^2}, \tag{2.33}$$

$$\mathbb{E}\sup_t|B^\circ(t)| = \sqrt{\frac{\pi}{2}}\log 2 = 0.868731\ldots, \tag{2.34}$$

as recorded, for example, in [7]. □

At $j = \lfloor m/2 \rfloor$, the ordinary hypergeometric central limit theorem gives, for every fixed $c > 0$,

$$\mathbb{P}\left(|U_{\lfloor m/2 \rfloor}| > c\sigma_m\sqrt{m}\right) \longrightarrow 2\overline{\Phi}(2c) > 0. \tag{2.35}$$

Equations (2.23) and (2.35) show that $m^{1/2}$ is the critical finite scale of the centered class. This is the same exponent that occurs in the classical RH criterion, but the logical statements are different.

Warning 2.5 (Endpoint and coupling). The deterministic endpoint satisfies

$$\left|\max_j |T_j| - \max_j |U_j|\right| \le |S|, \qquad \max_j |T_j| \ge |S|. \tag{2.36}$$

A concentration statement for U_j does not control $S = M(N)$. Moreover, the finite classes at different values of N are not a single canonically coupled random sequence. The law of the iterated logarithm is therefore not a prediction for one draw from $\Omega_N^{(1)}$; it concerns an almost-sure envelope along a coupled infinite sequence [10].

Warning 2.6 (Why typicality does not imply RH). At $N = 5\cdot 10^6$, a designated member has uniform mass

$$|\Omega_N^{(1)}|^{-1} = 10^{-915017.333\ldots}. \tag{2.37}$$

An event containing a $1 - o(1)$ fraction of a finite class need not contain that designated member. In addition, the class deliberately discards the divisor relations that define μ. No concentration inequality over the class transfers by itself to the Möbius word.

3. Normalized logarithmic quadratic path mass

3.1. **Definition and plateau representation.**

Fix $1 \le N_0 < N$ and put

$$L_N = \log(N/N_0). \tag{3.1}$$

Definition 3.1 (Normalized logarithmic quadratic path mass)**.** For a finite word w and its partial sums $R_w(n)$, define

$$\mathcal{Q}_{N_0,N}(w) = \frac{1}{L_N} \sum_{N_0 < n \le N} \frac{R_w(n)^2}{n^2}. \tag{3.2}$$

This is a positive logarithmically weighted second moment of the summatory path. No physical interpretation is assumed.

For the fixed mask, define the plateau weights

$$w_j = w_j^{(N_0,N)} = \sum_{\substack{N_0 < n \le N \\ Q(n) = j}} \frac{1}{n^2}, \qquad 1 \le j \le m. \tag{3.3}$$

Then the path functional is the quadratic form

$$L_N \mathcal{Q}_{N_0,N}(w) = \sum_{j=1}^{m} w_j T_j^2. \tag{3.4}$$

The weights depend only on the squarefree mask; all order dependence is in the bridge coordinates T_j.

3.2. **Relation to quadratic forms under permutation.** Let $\xi = (\xi_1, \dots, \xi_m)^{\mathsf{T}}$. Expanding the partial sums in (3.4) gives

$$L_N \mathcal{Q}_{N_0,N}(w) = \xi^{\mathsf{T}} \mathbf{K}_N \xi, \tag{3.5}$$

$$(\mathbf{K}_N)_{rs} = \sum_{j=\max(r,s)}^{m} w_j, \qquad 1 \le r, s \le m. \tag{3.6}$$

Thus the model is a quadratic form evaluated on a uniform permutation of a fixed multiset of signs. Permutation quadratic forms, quadratic-assignment statistics, and their exact moments have an established literature [12, 14, 28]. In particular, exact permutation moments of general quadratic forms are not claimed here as a new category of result.

The specific feature used below is the nested-max kernel (3.6). For this kernel and the fixed-count sign population, the covariance of squared prefix sums factorizes on the upper triangle, reducing the variance to one prefix pass; the same prefix geometry also yields the local transposition formula in Section 3.5. These special simplifications, rather than the generic quadratic-form representation, delimit the finite probabilistic contribution.

3.3. **Exact finite-class mean and variance.** Define two mask sums

$$\Sigma_1(N_0, N) = \sum_{N_0 < n \le N} \frac{Q(n)}{n^2} = \sum_{j=1}^{m} j w_j, \tag{3.7}$$

$$\Sigma_2(N_0, N) = \sum_{N_0 < n \le N} \frac{Q(n)^2}{n^2} = \sum_{j=1}^{m} j^2 w_j. \tag{3.8}$$

For $m \ge 2$, let

$$A = \frac{m^2 - S^2}{m(m-1)} = 1 - \mu_2, \qquad B = \frac{S^2 - m}{m(m-1)} = \mu_2. \tag{3.9}$$

Theorem 3.2 (Exact mean over the complete class)**.** *For the uniform measure on $\Omega_N^{(1)}$,*

$$\overline{\mathcal{Q}}_N := \mathbb{E}_\pi \mathcal{Q}_{N_0,N}(\pi) = \frac{A\Sigma_1 + B\Sigma_2}{L_N} \tag{3.10}$$

$$= \frac{1}{L_N} \sum_{j=1}^{m} w_j \{ j + \mu_2 j(j-1) \}. \tag{3.11}$$

Thus all cutoffs can be evaluated in one streaming pass, because Σ_1 and Σ_2 are cumulative while A and B depend only on the terminal values m and S.

Proof. Substitute (2.14) into (3.4). The identity $\mathbb{E}T_j^2 = Aj + Bj^2$ then gives (3.10); expanding $A = 1 - \mu_2$ gives (3.11). □

Theorem 3.3 (Exact variance in one prefix pass)**.** *Assume $m \ge 4$, and define*

$$P_k = \sum_{j<k} w_j \phi(j). \tag{3.12}$$

Then

$$\mathrm{Var}_\pi(\mathcal{Q}_{N_0,N}(\pi)) = \frac{1}{L_N^2} \sum_{k=1}^{m} w_k \psi(k) \{ w_k \phi(k) + 2P_k \}. \tag{3.13}$$

The formula uses $O(m)$ operations and $O(1)$ running state once the weights are streamed.

Proof. Expand the variance of $\sum_j w_j T_j^2$, use symmetry of covariance, and apply Theorem 2.2:

$$L_N^2 \operatorname{Var}(\mathcal{Q}) = \sum_{k=1}^{m} w_k^2 \phi(k)\psi(k) + 2 \sum_{1\le j<k\le m} w_j w_k \phi(j)\psi(k) \tag{3.14}$$

$$= \sum_{k=1}^{m} w_k \psi(k) \left\{ w_k \phi(k) + 2 \sum_{j<k} w_j \phi(j) \right\}. \tag{3.15}$$

This is (3.13). □

Proposition 3.4 (Semiseparable covariance and tridiagonal precision)**.** *Let $I = \{\ell, \ell+1, \dots, u\} \subseteq \{1, \dots, m\}$, and let $\mathbf{C}_I$ be the covariance matrix of $(T_j^2)_{j\in I}$. Assume that $\psi(j) \neq 0$ on I and that $\mathbf{C}_I$ is positive definite. Define*

$$b_j = \psi(j), \qquad r_j = \frac{\phi(j)}{\psi(j)}, qquad \delta_\ell = r_\ell, \qquad \delta_j = r_j - r_{j-1} \quad (j > \ell). \tag{3.16}$$

Then every δ_j is positive and

$$\mathbf{C}_I = \mathbf{D}\mathbf{L} \operatorname{diag}(\delta_\ell, \dots, \delta_u) \mathbf{L}^{\mathsf{T}} \mathbf{D}, \tag{3.17}$$

where $\mathbf{D} = \operatorname{diag}(b_\ell, \dots, b_u)$ and $L_{ij} = \mathbf{1}_{\{i\ge j\}}$. Consequently,

$$\mathbf{C}_I^{-1} = \mathbf{D}^{-1} \mathbf{H} \mathbf{D}^{-1}, \tag{3.18}$$

where $\mathbf{H}$ is tridiagonal with

$$H_{jj} = \delta_j^{-1} + \delta_{j+1}^{-1} \quad (j < u), \qquad H_{uu} = \delta_u^{-1}, \tag{3.19}$$

$$H_{j,j+1} = H_{j+1,j} = -\delta_{j+1}^{-1}. \tag{3.20}$$

Moreover,

$$\det \mathbf{C}_I = \left(\prod_{j\in I} b_j^2 \right) \left(\prod_{j\in I} \delta_j \right). \tag{3.21}$$

Thus solves and log-determinants for every such block require only $O(|I|)$ operations.

Proof. For $i \le j$, Theorem 2.2 gives

$$C_{ij} = \phi(i)\psi(j) = b_i b_j r_i. \tag{3.22}$$

The matrix with entries $r_{\min(i,j)}$ is $\mathbf{L} \operatorname{diag}(\delta_j) \mathbf{L}^{\mathsf{T}}$, proving (3.17). Positive definiteness and invertibility of $\mathbf{D}$ force $\delta_j > 0$. Since $\mathbf{L}^{-1}$ has ones on the diagonal and minus ones on the first subdiagonal, direct multiplication gives (3.18)–(3.20); taking determinants gives (3.21). □

Remark 3.5. The one-pair or Green-matrix algebra in Proposition 3.4 is classical. What is specific here is that the exact squared-prefix covariance generated by the fixed-count sign orbit has precisely this form. Tridiagonal inverse covariance is a Gaussian Markov property only for a Gaussian surrogate; no Markov claim is made for the non-Gaussian vector (T_j^2) itself.

Remark 3.6 (Exact formula versus numerical evaluation). Equations (3.10) and (3.13) are exact finite identities. Decimal values reported later are binary64 evaluations with explicit forward-error bounds; they are not directed-rounding interval evaluations.

3.4. **The exact order-dependent term.** The elementary square increment is

$$T_j^2 - T_{j-1}^2 = 1 + 2\xi_j T_{j-1}. \tag{3.23}$$

Define the future plateau weight

$$W_b = \sum_{j=b}^{m} w_j. \tag{3.24}$$

Proposition 3.7 (Diagonal term plus signed restoration)**.** *Every member of* $\Omega_N^{(1)}$ *satisfies*

$$L_N \mathcal{Q}_{N_0,N}(w) = \Sigma_1(N_0, N) + 2\mathcal{J}_N(w), \tag{3.25}$$

$$\mathcal{J}_N(w) = \sum_{b=2}^{m} W_b \xi_b T_{b-1}. \tag{3.26}$$

The first term is a mask invariant. Consequently the complete class is ordered by the single scalar $\mathcal{J}_N$*.*

Proof. Insert (3.23) into $T_j^2 = \sum_{b\le j}(T_b^2 - T_{b-1}^2)$, multiply by w_j, and interchange the two finite sums. The diagonal contribution is $\sum_j j w_j = \Sigma_1$. □

Exchangeability also gives

$$\mathbb{E}_\pi(\xi_b T_{b-1}) = (b-1)\mu_2, \tag{3.27}$$

and hence

$$\mathbb{E}_\pi \mathcal{J}_N = \mu_2 \sum_{b=2}^{m} (b-1) W_b. \tag{3.28}$$

For the nearly balanced classes in the data, μ_2 is close to $-1/(m-1)$. The much smaller observed value of $\mathcal{Q}$ therefore means that the ordered products $\xi_b T_{b-1}$, with their future weights, are substantially more negative than their exchangeable-class average.

3.5. **Exact change under a transposition.** Let $a < b$ be support indices carrying opposite signs. Swapping them preserves the mask, sign counts, and endpoint.

Proposition 3.8 (Opposite-sign transposition formula)**.** *If* $\xi_a = +1$ *and* $\xi_b = -1$*, then*

$$L_N\{\mathcal{Q}(\pi \circ \tau_{a,b}) - \mathcal{Q}(\pi)\} = 4\sum_{j=a}^{b-1} w_j(1 - T_j). \tag{3.29}$$

If $\xi_a = -1$ *and* $\xi_b = +1$*, then*

$$L_N\{\mathcal{Q}(\pi \circ \tau_{a,b}) - \mathcal{Q}(\pi)\} = 4\sum_{j=a}^{b-1} w_j(1 + T_j). \tag{3.30}$$

Proof. In the first case the path decreases by 2 on $a \le j < b$ and is unchanged elsewhere, so $(T_j - 2)^2 - T_j^2 = 4(1 - T_j)$. The second case is analogous. □

With

$$\overline{T}_w[a, b) = \frac{\sum_{j=a}^{b-1} w_j T_j}{\sum_{j=a}^{b-1} w_j}, \tag{3.31}$$

on intervals of positive total weight, a $+,-$ transposition decreases the quadratic mass exactly when $\overline{T}_w[a, b) > 1$, and a $-,+$ transposition decreases it exactly when $\overline{T}_w[a, b) < -1$. Thus a local minimum under all admissible transpositions is characterized by these two orientation-sensitive families of inequalities. They are not equivalent to a single condition $|\overline{T}_w[a, b)| \le 1$ over every interval, because not every interval has both endpoint orientations. If the total interval weight is zero, the mean in (3.31) is left undefined and the transposition is an exact tie.

The lower set determined by the observed word is therefore

$$\mathcal{A}_N = \{\pi \in \Omega_N^{(1)} : \mathcal{Q}_{N_0,N}(\pi) \le \mathcal{Q}_{N_0,N}(\mu)\} = \{\pi : \mathcal{J}_N(\pi) \le \mathcal{J}_N(\mu)\}. \tag{3.32}$$

3.6. **Discrete convex geometry and global minimizers.** Write

$$F_N(\xi) = L_N \mathcal{Q}_{N_0,N}(\xi) = \sum_{j=1}^{m} w_j T_j^2, \qquad r = \min(p,q) = \frac{m-|S|}{2}, \tag{3.33}$$

and define the endpoint-feasible lower envelope

$$h_j = \begin{cases} j \bmod 2, & 0 \le j \le 2r, \\ j - 2r, & 2r < j \le m. \end{cases} \tag{3.34}$$

Theorem 3.9 (Exact global minimum)**.** *For arbitrary weights $w_j \ge 0$,*

$$\min_{\xi:\, \#\{j:\xi_j=+1\}=p} F_N(\xi) = \sum_{j=1}^{m} w_j h_j^2. \tag{3.35}$$

A word is a minimizer if and only if

$$T_j^2 = h_j^2 \qquad \text{for every } j \text{ with } w_j > 0. \tag{3.36}$$

If $w_j > 0$ for every $1 \le j < m$, all minimizers consist of r independently chosen pairs $(+,-)$ or $(-,+)$, followed by $|S|$ copies of the majority sign. In that strictly positive case there are exactly 2^r minimizers.

Proof. At time j, every feasible path satisfies

$$T_j \equiv j \pmod 2, \qquad |T_j| \le j, \qquad |S - T_j| \le m - j. \tag{3.37}$$

For $j \le 2r = m - |S|$, parity forces $|T_j| \ge j \bmod 2$. If $S \ge 0$ and $j > 2r = m - S$, endpoint reachability gives $T_j \ge S - (m-j) = j - 2r$; applying the same argument to $-T_j$ handles $S < 0$. Hence $T_j^2 \ge h_j^2$ at every time. The lower envelope is attained by opposite-sign pairs for the first $2r$ steps and then by the majority sign, proving (3.35). Equality in a nonnegative weighted sum is equivalent to (3.36). When every interior weight is positive, the even times through $2r$ must all be returns to zero, so each consecutive pair is opposite; the endpoint then forces the remaining signs. This also gives the count 2^r. □

For the truncated functional used in the numerical work, some early plateau weights vanish. Equation (3.35) remains exact, but the count 2^r need not: zero-weight prefix coordinates may create additional minimizers. The distinction is necessary when $N_0 > 1$.

Set $x_i = (\xi_i + 1)/2$, so that $x_i \in \{0,1\}$, $\sum_i x_i = p$, and $X_j = \sum_{i \le j} x_i$. Then

$$F_N(x) = \sum_{j=1}^{m} w_j (2X_j - j)^2. \tag{3.38}$$

Proposition 3.10 (Murota M-convexity and local–global optimality)**.** *Extend* (3.38) *by $+\infty$ outside the binary fixed-cardinality domain. The resulting function is M-convex in the sense of discrete convex analysis. Consequently,*

$$\xi \text{ is a global minimizer} \iff \text{no opposite-sign transposition decreases } F_N. \tag{3.39}$$

Every sequence of strictly improving exchanges terminates at a global minimizer.

Proof. The prefix sets $\{1,\dots,j\}$, the singleton sets, and the full ground set form a laminar family. Attach to each singleton the convex indicator of $\{0,1\}$, and to the ground set the convex indicator of the value p. For the prefix terms,

$$f_j(z) = w_j(2z - j)^2, \qquad f_j(z+1) - 2f_j(z) + f_j(z-1) = 8w_j \ge 0. \tag{3.40}$$

The laminar convex construction therefore gives an M-convex function on the fixed-cardinality base family; see [19, Section 6.3]. The exchange optimality theorem for M-convex functions says that absence of an improving unit exchange is equivalent to global optimality. On this binary domain, unit exchanges are precisely opposite-sign transpositions. Strict descent must terminate because the domain is finite. □

The best improving exchange can itself be found in one pass. Define

$$A_k = \sum_{j=1}^{k} w_j(1 - T_j), \qquad B_k = \sum_{j=1}^{k} w_j(1 + T_j), \qquad A_0 = B_0 = 0. \tag{3.41}$$

For a $+,-$ pair $a < b$, the change is $4(A_{b-1} - A_{a-1})$; for a $-,+$ pair it is $4(B_{b-1} - B_{a-1})$. During a left-to-right scan, retain the largest previous A_{a-1} over positions with $\xi_a = +1$ and the largest previous B_{a-1} over positions with $\xi_a = -1$. Testing the compatible stored maximum at each right endpoint returns the best exchange in $O(m)$ time and $O(1)$ auxiliary memory. This proves the earlier algorithmic assertion; it does not bound the total number of descent steps.

Warning 3.11 (Two different meanings of convexity). The letter M in M-convexity refers to the terminology of discrete convex analysis, not to the Mertens function. The sequence $M(n)$ is neither discretely convex nor concave, since

$$\Delta^2 M(n) = \mu(n+2) - \mu(n+1) \tag{3.42}$$

takes both signs. It is not eventually convex or eventually concave: $\mu(n) \in \{-1, 0, 1\}$ would then be eventually monotone and hence eventually constant, whereas all three values occur arbitrarily far out. The stronger positive-lower-density statement for all nine adjacent pairs is proved in [17]. Already at the beginning,

$$\Delta^2 M(0) = \mu(2) - \mu(1) = -2, \qquad \Delta^2 M(2) = \mu(4) - \mu(3) = 1. \tag{3.43}$$

The step extension is discontinuous at infinitely many integers, while the polygonal extension has successive slopes $\mu(n)$; neither extension is convex, concave, eventually convex, or eventually concave. Even squaring the path does not repair discrete convexity, since the first three squared values are $0, 1, 0$. What is convex here is the quadratic functional of the complete path vector and, more specifically, its M-convex restriction to a fixed-count binary slice.

3.7. **Occupation interpretation.** Put $d = 1/\zeta(2) = 6/\pi^2$, the standard squarefree density [20]. The exact probability measure relevant to logarithmic occupation is

$$\nu_N(\{j\}) = \frac{jw_j}{D_N}, \qquad D_N = \sum_{j=1}^{m} jw_j = \Sigma_1(N_0, N). \tag{3.44}$$

For $X_j = T_j/\sqrt{j}$, its normalized quadratic occupation level is

$$A_N^{\mathrm{occ}} = \int X_j^2 \, d\nu_N(j) = \frac{L_N \mathcal{Q}_{N_0,N}(w)}{\Sigma_1(N_0, N)}. \tag{3.45}$$

The layer-cake identity gives the further exact representation

$$\mathcal{Q}_{N_0,N}(w) = \frac{\Sigma_1}{L_N} \int_0^\infty 2\lambda \, \nu_N\{j : |X_j| > \lambda\} \, d\lambda. \tag{3.46}$$

For every $\lambda > 0$, Markov's inequality gives the exact consequence

$$\nu_N\{j : |T_j| > \lambda\sqrt{j}\} \le \frac{L_N \mathcal{Q}_{N_0,N}(w)}{\lambda^2 \Sigma_1(N_0, N)}. \tag{3.47}$$

Proposition 3.12 (Aggregate logarithmic weighting)**.** *Uniformly for $N > N_0 \geq 1$,*

$$\Sigma_1(N_0, N) = d \log(N/N_0) + O(N_0^{-1/2}). \tag{3.48}$$

Moreover, if

$$\eta_{N_0,N} = \frac{1}{\Sigma_1} \sum_{N_0 < n \leq N} \frac{Q(n)}{n^2} \delta_{\log(n/N_0)/L_N}, \tag{3.49}$$

then $\eta_{N_0,N}$ converges weakly to Lebesgue probability measure on $[0,1]$ whenever $L_N \to \infty$. The measure in (3.49) *is the push-forward of ν_N after resolving each plateau into its integer positions.*

Proof. The elementary estimate $Q(n) = dn + O(\sqrt{n})$ gives

$$\begin{aligned} \Sigma_1(N_0, N) &= d \sum_{N_0 < n \leq N} \frac{1}{n} + O\left(\sum_{n > N_0} n^{-3/2} \right) \\ &= dL_N + O(N_0^{-1/2}), \end{aligned} \tag{3.50}$$

which proves (3.48). Against a continuous test function on $[0,1]$, the main term is a Riemann sum for uniform measure in the variable $u = \log(n/N_0)/L_N$. The total squarefree-counting error is $O(N_0^{-1/2})$; after division by $\Sigma_1 \asymp L_N$, it vanishes. Finally, $jw_j = \sum_{Q(n)=j} Q(n)/n^2$, which proves the push-forward statement. □

There is deliberately no pointwise claim $w_j \sim d/j^2$: an individual plateau weight also contains the fluctuating gap to the next support position. The rigorous statement is aggregate. In particular,

$$\mathcal{Q}_{N_0,N}(w) = \frac{\Sigma_1}{L_N} \left\langle \frac{T_j^2}{j} \right\rangle_{\nu_N} = d \left\langle \frac{T_j^2}{j} \right\rangle_{\nu_N} + O(N_0^{-1/2}/L_N) \left\langle \frac{T_j^2}{j} \right\rangle_{\nu_N}. \tag{3.51}$$

At the complete cutoff, replacing the exact factor L_N/Σ_1 in (3.47) by its limiting value $\zeta(2)$ gives the descriptive levels 95.2 percent below $\sqrt{j}$ and 98.8 percent below $2\sqrt{j}$. These two percentages are asymptotic-density approximations; the exact finite statements use (3.47) with the computed Σ_1.

3.8. **A rigorous proportional-window limit.** The fixed lower cutoff used in the computation is singular from the functional-limit viewpoint. A clean distributional theorem is available when the observed integer window remains a fixed proportion of the full range.

Theorem 3.13 (Proportional-window Gaussian quadratic limit)**.** *Fix $0 < \alpha < 1$, put $N_0(N) = \lfloor \alpha N \rfloor$, and let $m_N = Q(N)$. Suppose that the fixed endpoint satisfies*

$$\frac{S_N}{\sqrt{m_N}} \longrightarrow y \in \mathbb{R}. \tag{3.52}$$

Under the uniform fixed-count orbit at cutoff N,

$$\mathcal{Q}_{\lfloor \alpha N \rfloor, N} \Longrightarrow \frac{d}{\log(1/\alpha)} \int_\alpha^1 \frac{\{B^\circ(t) + yt\}^2}{t^2}\, dt, \qquad d = \frac{1}{\zeta(2)}. \tag{3.53}$$

Proof. Uniformly for $t \in [\alpha, 1]$, the squarefree estimate gives

$$\frac{Q(\lfloor Nt \rfloor)}{Q(N)} \longrightarrow t, \qquad \frac{Q(N)}{N} \longrightarrow d. \tag{3.54}$$

Proposition 2.4, the deterministic time change in (3.54), and (3.52) imply

$$\frac{T_{Q(\lfloor Nt \rfloor)}}{\sqrt{Q(N)}} \Longrightarrow B^\circ(t) + yt \quad \text{in } C[\alpha, 1]. \tag{3.55}$$

Since t^{-2} is bounded on this interval,

$$\sum_{\lfloor \alpha N\rfloor < n \le N} \frac{T_{Q(n)}^2}{n^2} = \frac{Q(N)}{N}\,\frac{1}{N} \sum_{\lfloor \alpha N\rfloor < n \le N} \frac{\{T_{Q(n)}/\sqrt{Q(N)}\}^2}{(n/N)^2}$$

$$\Longrightarrow d \int_\alpha^1 \frac{\{B^\circ(t) + yt\}^2}{t^2}\, dt. \tag{3.56}$$

Finally, $\log\{N/\lfloor \alpha N\rfloor\} \to \log(1/\alpha)$. □

Proposition 3.14 (Explicit spectrum in the centered case)**.** *Let*

$$Z_\alpha = \int_\alpha^1 \frac{B^\circ(t)^2}{t^2}\, dt. \tag{3.57}$$

Then

$$Z_\alpha \overset{d}{=} \sum_{k\ge 1} \lambda_k Z_k^2, \qquad Z_k \overset{\text{iid}}{\sim} N(0,1), \qquad \lambda_k = \frac{1}{\beta_k^2 + 1/4}, \tag{3.58}$$

where the positive roots β_k are ordered increasingly and satisfy

$$\beta_k \cot(\beta_k \log(1/\alpha)) = -\frac{1}{2}. \tag{3.59}$$

For $s \ge 0$,

$$\mathbb{E}e^{-sZ_\alpha} = \prod_{k\ge 1} (1 + 2s\lambda_k)^{-1/2}, \tag{3.60}$$

$$\mathbb{E}Z_\alpha = \log(1/\alpha) - (1-\alpha), \tag{3.61}$$

$$\mathrm{Var}(Z_\alpha) = 2\sum_{k\ge 1} \lambda_k^2. \tag{3.62}$$

Proof. The covariance operator of B° in $L^2([\alpha,1], t^{-2}dt)$ has kernel $K(s,t) = \min(s,t) - st$. If f is an eigenfunction with eigenvalue λ, differentiating its integral equation twice yields

$$f''(t) + \frac{1}{\lambda t^2} f(t) = 0, \qquad f(1) = 0, \qquad f'(\alpha) = \frac{f(\alpha)}{\alpha}. \tag{3.63}$$

Writing $\lambda = (\beta^2 + 1/4)^{-1}$, the first boundary condition gives $f(t) = c\sqrt{t}\sin(\beta \log t)$. The second is exactly (3.59). The Karhunen–Loève expansion then proves (3.58) and (3.60). Finally,

$$\mathbb{E}Z_\alpha = \int_\alpha^1 \frac{t - t^2}{t^2}\, dt = \log(1/\alpha) - (1-\alpha), \tag{3.64}$$

and the variance of a centered Gaussian quadratic form is twice the squared Hilbert–Schmidt norm, giving (3.62). □

Theorem 3.13 is a genuine asymptotic theorem for the permutation model. It neither covers the computational regime with fixed $N_0 = 1000$, where $\alpha \downarrow 0$, nor supplies exponential equivalence or a large-deviation principle. In particular, it does not upgrade the Ornstein–Uhlenbeck lower-tail heuristic in Section 5.2 to a theorem.

3.9. **Support-time coefficients and a feedback heuristic.** For an interval $u < j \le v$, define

$$c_{u,v} = \frac{1}{v-u} \sum_{j=u+1}^{v} \frac{T_j^2}{j}. \tag{3.65}$$

Under an explicitly specified ensemble, one may also define

$$c(K) = \frac{\mathrm{Var}(T_K)}{K}. \tag{3.66}$$

These are distinct quantities. The algebraic normalization

$$a_N = \zeta(2)\mathcal{Q}_{N_0,N}(\mu) \tag{3.67}$$

is not an independent measurement of either coefficient.

Heuristic 3.15 (Critical restoring drift)**.** This is a heuristic argument. Suppose a process obeys

$$\mathbb{E}(\xi_{j+1} \mid T_j) \approx -\frac{2a}{j}T_j. \tag{3.68}$$

Writing $V_j = \mathbb{E}T_j^2$ gives

$$V_{j+1} - V_j \approx 1 - \frac{4a}{j}V_j. \tag{3.69}$$

A solution $V_j \sim cj$ then satisfies

$$c = \frac{1}{1+4a}, \qquad a = \frac{c^{-1}-1}{4}. \tag{3.70}$$

The power j^{-1} is critical: a drift $j^{-\gamma}$ formally gives $V_j/j \to 1$ for $\gamma > 1$, $V_j/j \to 0$ for $\gamma < 1$, and permits a strict constant in $(0,1)$ only for $\gamma = 1$. This model can explain how a cumulative suppression coexists with a null nearest-neighbour persistence statistic, but (3.70) is not an identity for the Möbius word.

The scale measurements in Table 3 fluctuate between approximately 0.036 and 0.068, without monotone convergence. A sinusoidal fit in $\log_2 K$ is reported in Appendix A; its out-of-sample performance does not establish a predictive law.

4. Excursion degeneracy and exact return counts

4.1. **Reflection orbit.** Let

$$0 = \tau_0 < \tau_1 < \cdots < \tau_K \tag{4.1}$$

be the successive completed zero-to-zero excursions of the support-time path. Write the r-th completed excursion as

$$T|_{[\tau_{r-1},\tau_r]} = \varepsilon_r H_r, \qquad \varepsilon_r \in \{-1,+1\}, \qquad H_r \geq 0. \tag{4.2}$$

Proposition 4.1 (Pointwise absolute-path orbit)**.** *Reflecting any subset of the K completed excursions preserves the zero mask, the two sign counts, the endpoint, and every value $R_w(n)^2$. Moreover, the resulting orbit exhausts the pointwise absolute-path class:*

$$\mathcal{O}_N(\mu) = \Omega_N^{(2)}, \qquad |\mathcal{O}_N(\mu)| = |\Omega_N^{(2)}| = 2^K, \qquad \mathcal{Q}_{N_0,N}(w) = \mathcal{Q}_{N_0,N}(\mu) \quad (w \in \mathcal{O}_N(\mu)). \tag{4.3}$$

If $M(N) \neq 0$, the terminal incomplete excursion after the last return is not reflected, because its reflection would change the fixed endpoint.

Proof. A completed excursion contains equally many positive and negative steps. Reflection changes every sign in that segment, preserves those counts, and replaces the path by its negative only within the segment. Independent choices therefore give 2^K distinct words in $\Omega_N^{(2)}$.

Conversely, take $w \in \Omega_N^{(2)}$. Between two consecutive zeros τ_{r-1} and τ_r, the reference path has no interior zero. Because both paths have unit increments in support time and the same absolute value at every support index, the sign relating them cannot change inside that interval. Hence the restriction of w to the excursion is either the original sign block or its complete reflection. These alternatives are independent across completed excursions. On a terminal incomplete segment, the fixed endpoint $T_m = S \neq 0$ forces the original orientation. Thus there are no further elements of $\Omega_N^{(2)}$, proving exhaustivity and the asserted equality. When $S = 0$, the terminal segment is itself the last completed excursion. □

Proposition 4.2 (Sharp quadratic content of one excursion)**.** *Let $H_0 = H_{2\ell} = 0$, let $H_t > 0$ for $1 \le t < 2\ell$, and assume $|H_t - H_{t-1}| = 1$. Then*

$$5\ell - 4 \le \sum_{t=1}^{2\ell} H_t^2 \le \ell^2 + \frac{(\ell-1)\ell(2\ell-1)}{3}. \tag{4.4}$$

The lower equality path is

$$1, 2, 1, 2, \ldots, 1, 0, \tag{4.5}$$

and the upper equality path is

$$1, 2, \ldots, \ell, \ell-1, \ldots, 1, 0. \tag{4.6}$$

Both are unique among positive excursions. Their negatives give the same quadratic content.

Proof. Strict positivity and parity give $H_t \ge 1$ at odd interior times and $H_t \ge 2$ at even interior times. Summing their squares gives $\ell \cdot 1^2 + (\ell-1)\cdot 2^2 = 5\ell - 4$, attained by (4.5). Endpoint reachability gives $H_t \le \min(t, 2\ell - t)$. The sum of the squared upper envelope is

$$\ell^2 + 2\sum_{t=1}^{\ell-1} t^2 = \ell^2 + \frac{(\ell-1)\ell(2\ell-1)}{3}, \tag{4.7}$$

attained by (4.6). Equality at every time forces the displayed paths. □

For nonconstant plateau weights, Proposition 4.2 does not by itself determine the contribution to F_N. If $w_{\min}$ and $w_{\max}$ are the smallest and largest weights across that excursion, it gives the valid deterministic bounds

$$w_{\min}(5\ell - 4) \le \sum_{t=1}^{2\ell} w_{\tau+t} H_t^2 \le w_{\max}\left\{\ell^2 + \frac{(\ell-1)\ell(2\ell-1)}{3}\right\}. \tag{4.8}$$

Remark 4.3 (Consequence for selection)*.* The reflection orbit describes degeneracy after the absolute path has already been fixed. It does not explain why the observed absolute path has small quadratic mass. It proves instead that no functional depending only on $R_w(n)^2$ can select a unique orientation.

4.2. **The quadratic profile and the absolute-path quotient.** A single terminal value of $\mathcal{Q}$ compresses the path to one number. The complete cumulative profile loses exactly the orientation information and no more. Define

$$\mathscr{A}_x(w) = \sum_{n \le x} \frac{R_w(n)^2}{n^2}, \qquad 1 \le x \le N. \tag{4.9}$$

Proposition 4.4 (Exact information content of the quadratic profile)**.** *For any two words w, v on $[1, N]$,*

$$\mathscr{A}_x(w) = \mathscr{A}_x(v) \quad (1 \le x \le N) \quad \Longleftrightarrow \quad |R_w(x)| = |R_v(x)| \quad (1 \le x \le N). \tag{4.10}$$

Equivalently, the full cumulative quadratic profile is a complete invariant of the absolute path. Within $\Omega_N^{(1)}$, its fiber through μ is the reflection orbit $\mathcal{O}_N(\mu)$ of Proposition 4.1.

Proof. The successive difference is

$$\mathscr{A}_x(w) - \mathscr{A}_{x-1}(w) = \frac{R_w(x)^2}{x^2}. \tag{4.11}$$

Thus equality of the profiles gives equality of every squared path value; the converse follows directly from (4.9). The last assertion is exactly the exhaustivity statement in Proposition 4.1. □

Define

$$w \sim_{\mathrm{abs}} v \quad \Longleftrightarrow \quad |R_w(x)| = |R_v(x)| \quad (1 \le x \le N). \tag{4.12}$$

The appropriate object for any collection of even path functionals is therefore the quotient

$$\Omega_N^{(1)} / \sim_{\mathrm{abs}} . \tag{4.13}$$

The selection problem separates into two logically distinct layers:

$$\text{Möbius selection} = \text{absolute-path selection} + \text{arithmetic orientation selection}. \tag{4.14}$$

The quadratic mass and its multiscale profile address the first layer. An observable for the second layer must change under at least one excursion reflection and must be specified independently of the observed orientations.

4.3. **Expected number of returns.** For $w \in \Omega_N^{(1)}$, define

$$Z_N(w) = \#\{1 \le j \le m : T_j(w) = 0\}. \tag{4.15}$$

Proposition 4.5 (Exact first moment of the return count)**.** *Under the uniform fixed-count measure,*

$$\mathbb{E}_\pi Z_N = \sum_{k=1}^{\min(p,q)} \frac{\binom{p}{k}\binom{q}{k}}{\binom{m}{2k}}. \tag{4.16}$$

Proof. A return is possible only at time $2k$. Among the $\binom{m}{2k}$ equally likely subsets of sampled signs, precisely $\binom{p}{k}\binom{q}{k}$ contain k signs of each type. Sum the indicators of the return events. □

At $N = 5 \cdot 10^6$, the exact and observed values reported in Appendix A are

$$\mathbb{E}_\pi Z_N = 1623.0496, \qquad Z_N(\mu) = 6016, \qquad \frac{Z_N(\mu)}{\mathbb{E}_\pi Z_N} = 3.7066. \tag{4.17}$$

The same path has longest completed-excursion fraction 0.11737, compared with a reported bridge-sample mean 0.5301. These are related descriptions of a path spending unusually little logarithmic mass far from zero; they are not independent events whose probabilities may be multiplied.

5. Finite placement of the Möbius word

Define the class-relative ratio and standardized displacement

$$\rho_Q(N) = \frac{\mathcal{Q}_{N_0,N}(\mu)}{\overline{\mathcal{Q}}_N}, \qquad Z_Q(N) = \frac{\mathcal{Q}_{N_0,N}(\mu) - \overline{\mathcal{Q}}_N}{\sqrt{\mathrm{Var}_\pi(\mathcal{Q}_{N_0,N})}}. \tag{5.1}$$

Proposition 5.1 (Finite separation)**.** *Let the input be the signed-byte file with 455,052,511 entries and SHA-256 digest*

$$\texttt{0e551f47d924ef6f3e348e131fdf53e0c677ad5e5471ef9d6249b3fd662e2ea4}. \tag{5.2}$$

Under the binary64 and execution assumptions specified in Appendix B, the fourteen cutoffs in Table 1 satisfy

$$\mathcal{Q}_{1000,N}(\mu) < \overline{\mathcal{Q}}_N. \tag{5.3}$$

At the complete cutoff,

$$\mathcal{Q}_{1000,455052511}(\mu) = 0.029334, \quad \overline{\mathcal{Q}}_{455052511} = 0.56628, \quad \rho_Q(455052511) = 0.0518. \tag{5.4}$$

At $N = 5 \cdot 10^6$, the higher-precision reported values are

$$\mathcal{Q}_{1000,N}(\mu) = 0.0295150019629, \quad \overline{\mathcal{Q}}_N = 0.548481916755, \quad \rho_Q(N) = 0.0538121696655. \tag{5.5}$$

Proof. The class mean is evaluated from Theorem 3.2; the observed value is a positive weighted sum. The streaming implementation uses blocked summation, compensation between blocks, and an a-priori forward-error propagation under the assumptions in Appendix B. For each cutoff the upper endpoint of the propagated interval for $\mathcal{Q}(\mu) - \overline{\mathcal{Q}}_N$ is negative. The smallest ratio of the absolute separation to the propagated bound is $5.75 \cdot 10^{10}$. Division gives (5.4) and (5.5). The proposition is conditional on the identified file and execution model; the exact symbolic identities do not depend on either. □

Remark 5.2 (What has been proved). Proposition 5.1 proves comparison with the exact class mean at fourteen finite cutoffs. It does not prove that $\rho_Q(N)$ is always near 0.05, that the ratio is monotone, or that it has a limit. Indeed the first tabulated value is 0.082224, and the later values exhibit small nonmonotone fluctuations.

At the complete cutoff,

$$m = 276{,}638{,}778, \qquad S = -5414, \qquad \max_{n \le N} |M(n)| = 6695. \tag{5.6}$$

Since $\sqrt{m} = 16632$ to the displayed precision, the observed maximum is

$$\frac{6695}{\sqrt{m}} = 0.402, \qquad \frac{6695}{0.868731\sqrt{m}} = 0.463. \tag{5.7}$$

The second denominator is the limiting mean maximum of a nearly balanced finite bridge. This is a finite comparison; it is not a law-of-the-iterated- logarithm statement.

5.1. **Lower-tail sampling.** The exact lower-tail fraction is

$$P_Q(N) = \frac{\#\{\pi \in \Omega_N^{(1)} : \mathcal{Q}_{N_0,N}(\pi) \le \mathcal{Q}_{N_0,N}(\mu)\}}{|\Omega_N^{(1)}|}. \tag{5.8}$$

No closed form for (5.8) is established here. Uniform fixed-count sampling reported 412 hits in 300,000 draws at $N = 10^4$, 15 hits in 150,000 draws at $N = 3 \cdot 10^4$, and zero hits at each of the four larger sampled cutoffs. Full results and exact zero-hit bounds appear in Table 2.

Experiment 5.3 (Failure of a two-moment tail approximation)**.** A scaled chi-square law fitted to the exact mean and variance predicts, across the six cutoffs, expected hit counts

$$(20772,\ 6113.7,\ 2218.2,\ 477.88,\ 61.85,\ 23.242), \tag{5.9}$$

whereas the observed counts are

$$(412,\ 15,\ 0,\ 0,\ 0,\ 0). \tag{5.10}$$

At $N = 5 \cdot 10^6$, the approximation gives

$$P_Q^{(\chi^2)} = 0.0023242189, \qquad \mathbb{P}_{\chi^2}(H = 0) = 7.8397 \cdot 10^{-11}. \tag{5.11}$$

For each row, using the fitted chi-square success probability in the corresponding binomial model, the explicitly used lower-tail probability $\mathbb{P}_{\chi^2}(H \le H_{\mathrm{obs}})$ is below $7.84 \cdot 10^{-11}$. Thus the fitted two-moment proxy is decisively incompatible with the sampled lower tail; the statement is not a rejection of every distribution sharing those two moments.

Warning 5.4 (Mean separation is not a percentile). The ratio ρ_Q compares the observed value with the exact mean; it is not P_Q. Likewise, zero hits in R draws yield the exact one-sided 95-percent bound $P_Q < 1 - 0.05^{1/R}$, not an estimate of a probability of order 10^{-9}. At $N = 5 \cdot 10^6$, the class size is approximately $10^{915017.333}$, so exhaustive enumeration is impossible.

5.2. **Continuum diagnostics.** Put

$$d = \frac{1}{\zeta(2)}, \qquad y_N^2 = \frac{S^2}{m}, \qquad x_0 = e^{-L_N}. \tag{5.12}$$

Heuristic 5.5 (Finite-span Gaussian bridge)**.** This is a heuristic continuum replacement. Replacing the squarefree mask by density d and the hypergeometric bridge by a Gaussian bridge gives

$$\overline{\mathcal{Q}}_N^{\mathrm{cont}} = d\left[1 - (1-y_N^2)\frac{1-e^{-L_N}}{L_N}\right], \tag{5.13}$$

$$(\sigma_N^{\mathrm{cont}})^2 = \frac{d^2}{L_N^2}\Big[4L_N - 10 + 8(L_N+1)e^{-L_N} + 2e^{-2L_N} + 4y_N^2\{1 - e^{-2L_N} - 2L_N e^{-L_N}\}\Big]. \tag{5.14}$$

Dropping only the exponentially small term in the mean gives

$$\overline{\mathcal{Q}}_N^{\mathrm{principal}} = d\left[1 - \frac{1-y_N^2}{L_N}\right]. \tag{5.15}$$

When $y_N^2 = o(1)$,

$$\overline{\mathcal{Q}}_N^{\mathrm{cont}} = d\left(1 - \frac{1}{L_N}\right) + o(L_N^{-1}), \qquad (\sigma_N^{\mathrm{cont}})^2 = \frac{4d^2(L_N - 5/2)}{L_N^2} + o(L_N^{-2}). \tag{5.16}$$

At the complete cutoff, (5.15) gives 0.5662087354, while (5.13) gives 0.5662088271. The exact finite-class mean is 0.56628. The exact and continuum standard deviations are 0.30436 and 0.30433, respectively. These close numerical agreements are consistent with the approximation at the measured scale but do not prove a discrete-to-continuum error term.

Heuristic 5.6 (Ornstein–Uhlenbeck lower-tail rate)**.** The Lamperti transform

$$V(t) = e^{t/2}W(e^{-t}), \qquad \mathrm{Cov}(V(s), V(t)) = e^{-|t-s|/2} \tag{5.17}$$

is a stationary Ornstein–Uhlenbeck process with unit variance. The Feynman–Kac calculation within the Donsker–Varadhan framework [6] gives

$$\Lambda(\lambda) = \frac{1}{2}\left(\frac{1}{2} - \sqrt{\frac{1}{4} - 2\lambda}\right), \qquad \lambda < \frac{1}{8}, \tag{5.18}$$

$$I(a) = \sup_{\lambda < 1/8}\{\lambda a - \Lambda(\lambda)\} = \frac{(a-1)^2}{8a}. \tag{5.19}$$

Transferring this rate to the finite fixed-mask bridges requires an unproved large-deviation invariance principle and a boundary correction. The resulting prediction is

$$\log P_Q(N) \approx -L_N I(a_N) + \log \kappa(a_N, L_N), \qquad a_N = \zeta(2)\mathcal{Q}_{N_0,N}(\mu), \tag{5.20}$$

where the prefactor κ is not determined.

At $N = 5 \cdot 10^6$, the leading rate gives

$$a_N = 0.0485502, \qquad I(a_N) = 2.3307, \qquad \exp\{-L_N I(a_N)\} = 2.39 \cdot 10^{-9}. \tag{5.21}$$

At the complete cutoff the corresponding values are

$$a_N = 0.048252, \qquad I(a_N) = 2.3466, \qquad \exp\{-L_N I(a_N)\} = 5.28 \cdot 10^{-14}. \tag{5.22}$$

These are heuristic extrapolations far below the resolution of the uniform sampler. The straight line through the two cutoffs having positive hit counts has slope -2.38467; with only two points it carries no residual regression degrees of freedom. A descriptive six-cutoff binomial likelihood fit gives exponent 2.45306 and profile interval $[2.04052, 2.94753]$.

6. Classical analytic meaning of the quadratic scale

6.1. **Full-prefix and truncated normalizations.** The classical full-prefix quantity is

$$\mathfrak{Q}(N) = \frac{1}{\log N} \sum_{n \le N} \frac{M(n)^2}{n^2}. \tag{6.1}$$

For fixed N_0, set

$$K_{N_0} = \sum_{n \le N_0} \frac{M(n)^2}{n^2}. \tag{6.2}$$

Then the exact conversion is

$$\mathfrak{Q}(N) = \frac{K_{N_0} + L_N \mathcal{Q}_{N_0,N}(\mu)}{\log N}. \tag{6.3}$$

The two normalizations have the same limit whenever either limit exists, but they can be far apart at finite N. With $N_0 = 1000$,

$$K_{1000} = 2.068991, \qquad \mathfrak{Q}(455052511) = 0.122952, \tag{6.4}$$

whereas the truncated value is 0.029334. Thus the observed flatness of the latter is a property of the fixed lower cutoff, not evidence that the full-prefix normalization has numerically converged.

6.2. **Weak Mertens and RH.** For integer N,

$$\int_1^{N+1} \left(\frac{M(x)}{x}\right)^2 dx = \sum_{n=1}^{N} \frac{M(n)^2}{n(n+1)} = \sum_{n=1}^{N} \frac{n}{n+1} \frac{M(n)^2}{n^2}, \qquad \frac{1}{2} \le \frac{n}{n+1} < 1. \tag{6.5}$$

Hence boundedness of $\mathfrak{Q}(N)$ is equivalent, up to absolute comparison constants, to the weak Mertens conjecture

$$\int_1^X \left(\frac{M(x)}{x}\right)^2 dx \ll \log X. \tag{6.6}$$

It is distinct from the pointwise Mertens conjecture, which was disproved in [22]. This conjecture implies RH and simplicity of all nontrivial zeros; see Titchmarsh, pp. 376–380, and [21]. RH alone is not known to imply (6.6).

There is also a direct positive reformulation. Define

$$D(N) = \sum_{k \le N} \frac{\mu(k)^2}{k}, \tag{6.7}$$

$$C(N) = \sum_{k \le N} \frac{\mu(k) M(k-1)}{k}, \tag{6.8}$$

$$B(N) = D(N) + 2C(N). \tag{6.9}$$

Discrete summation by parts gives

$$B(N) = \frac{M(N)^2}{N} + \sum_{k<N} \frac{M(k)^2}{k(k+1)} \ge 0. \tag{6.10}$$

For any finite word w, write the corresponding positive path functional as

$$\mathcal{B}_N(w) = \frac{R_w(N)^2}{N} + \sum_{k<N} \frac{R_w(k)^2}{k(k+1)}. \tag{6.11}$$

Thus $B(N) = \mathcal{B}_N(\mu)$. It follows directly from the classical Mertens criterion that

$$\text{RH} \quad \Longleftrightarrow \quad B(N) \ll_\varepsilon N^\varepsilon \quad \text{for every } \varepsilon > 0. \tag{6.12}$$

This is a reformulation, not a new criterion for RH.

6.3. **An orbit-normalized RH criterion.** The permutation model supplies canonical denominators for (6.12). First use the uniform measure on the fixed-mask class $\Omega_N^{(0)}$, where the nonzero signs are independent and unbiased. Define

$$b_N^{(0)} = \mathbb{E}_0 \mathcal{B}_N(w). \tag{6.13}$$

For the fixed-count class, center the hypergeometric bridge at its deterministic linear interpolation:

$$U_{\pi,N}(n) = R_\pi(n) - \frac{Q(n)}{m} S, \qquad m = Q(N), \quad S = M(N), \tag{6.14}$$

and put

$$b_N^{(1),\circ} = 1 + \mathbb{E}_1 \sum_{n<N} \frac{U_{\pi,N}(n)^2}{n(n+1)}. \tag{6.15}$$

The added constant only prevents a degenerate denominator at the smallest cutoffs and has no asymptotic effect.

Theorem 6.1 (Exact orbit scales and their asymptotics)**.** *Let $d = 1/\zeta(2)$ and $m \geq 2$. The two orbit scales satisfy the exact formulae*

$$b_N^{(0)} = \frac{Q(N)}{N} + \sum_{n<N} \frac{Q(n)}{n(n+1)}, \tag{6.16}$$

$$b_N^{(1),\circ} = 1 + \frac{1 - S^2/m^2}{m-1} \sum_{n<N} \frac{Q(n)\{m - Q(n)\}}{n(n+1)}. \tag{6.17}$$

As $N \to \infty$,

$$b_N^{(0)} = d \log N + O(1), \qquad b_N^{(1),\circ} \sim d \log N. \tag{6.18}$$

Proof. Under $\Omega_N^{(0)}$, $\mathbb{E}_0 R_w(n)^2 = Q(n)$, which gives (6.16). Under $\Omega_N^{(1)}$, the finite-population variance formula gives

$$\mathbb{E}_1 U_{\pi,N}(n)^2 = \left(1 - \frac{S^2}{m^2}\right) \frac{Q(n)\{m - Q(n)\}}{m-1}, \tag{6.19}$$

which proves (6.17).

The elementary squarefree estimate $Q(n) = dn + O(\sqrt{n})$ yields

$$\sum_{n<N} \frac{Q(n)}{n(n+1)} = d \log N + O(1), \tag{6.20}$$

$$\sum_{n<N} \frac{Q(n)^2}{n(n+1)} = d^2 N + O(\sqrt{N}). \tag{6.21}$$

Since $m = dN + O(\sqrt{N})$, the sum in (6.17), divided by $m-1$, equals $d \log N + O(1)$. Finally, the classical estimate $M(N) = o(N)$ implies $S/m = o(1)$, proving (6.18). □

Define the two class-relative positive ratios

$$\mathscr{R}_0(N) = \frac{\mathcal{B}_N(\mu)}{b_N^{(0)}}, \qquad \mathscr{R}_1(N) = \frac{\mathcal{B}_N(\mu)}{b_N^{(1),\circ}}. \tag{6.22}$$

Theorem 6.2 (Orbit-normalized RH equivalence)**.** *The following statements are equivalent:*

$$\text{RH}; \tag{6.23}$$

$$\mathscr{R}_0(N) \ll_\varepsilon N^\varepsilon \quad \textit{for every } \varepsilon > 0; \tag{6.24}$$

$$\mathscr{R}_1(N) \ll_\varepsilon N^\varepsilon \quad \textit{for every } \varepsilon > 0. \tag{6.25}$$

Proof. By Theorem 6.1, both denominators in (6.22) are asymptotic to $d \log N$. Multiplying or dividing by a quantity of order $\log N$ does not change the collection of bounds $O_\varepsilon(N^\varepsilon)$. The result therefore follows from (6.12). □

Corollary 6.3 (Orbital growth exponent)**.** *Define*

$$\kappa_\Omega(\mu) = \limsup_{N\to\infty} \frac{\log\{1 + \mathscr{R}_1(N)\}}{\log N}. \tag{6.26}$$

Then

$$\mathrm{RH} \quad \Longleftrightarrow \quad \kappa_\Omega(\mu) = 0. \tag{6.27}$$

For comparison across general projectively specified words on the same support, define the endpoint-free exponent

$$\kappa_0(w) = \limsup_{N\to\infty} \frac{\log\{1 + \mathcal{B}_N(w)/b_N^{(0)}\}}{\log N}. \tag{6.28}$$

Because $b_N^{(0)} \sim b_N^{(1),\circ} \sim d \log N$,

$$\kappa_0(\mu) = \kappa_\Omega(\mu). \tag{6.29}$$

Remark 6.4 (What the equivalence does and does not say). For a general projectively specified infinite word w, the analogous condition $\kappa_0(w) = 0$ says only that its partial-sum path has the subpolynomial positive-functional growth associated with the square-root scale. Many deterministic and random words have that property. The equivalence with RH occurs only after setting $w = \mu$, because the classical Mertens criterion connects that particular coefficient sequence to $1/\zeta(s)$. Thus (6.27) classifies the growth of the distinguished word; it does not characterize μ among all words.

Remark 6.5 (A hierarchy of analytic strength). The orbit scale separates three statements:

$$\kappa_\Omega(\mu) > 0 \quad \Longleftrightarrow \quad \text{RH fails}, \tag{6.30}$$

$$\kappa_\Omega(\mu) = 0 \quad \Longleftrightarrow \quad \text{RH holds}, \tag{6.31}$$

$$\sup_N \mathscr{R}_1(N) < \infty \quad \Longleftrightarrow \quad \mathcal{B}_N(\mu) = O(\log N). \tag{6.32}$$

The last condition implies RH and is not known to follow from RH. It is a stronger mean-square-and-endpoint regime, not another formulation of RH.

Proposition 6.6 (Conditional meaning of a bounded class ratio)**.** *Fix N_0, and assume*

$$\frac{M(N)^2}{Q(N)\log(N/N_0)} \longrightarrow 0, \tag{6.33}$$

equivalently $M(N) = o(\sqrt{N \log N})$, using $Q(N) \sim dN$. Then

$$\overline{\mathcal{Q}}_N \longrightarrow d = \frac{1}{\zeta(2)} \tag{6.34}$$

and

$$\sup_{N\ge N_1} \rho_Q(N) < \infty \quad \Longleftrightarrow \quad \mathcal{Q}_{N_0,N}(\mu) = O(1) \quad \Longleftrightarrow \quad \mathfrak{Q}(N) = O(1). \tag{6.35}$$

Proof. The elementary estimate $Q(n) = dn + O(\sqrt{n})$ gives $\Sigma_1 = dL_N + O(1)$. In (3.11), the bridge correction is $O(L_N^{-1})$, while the endpoint contribution is $O(M(N)^2/[Q(N)L_N])$. Equation (6.33) proves (6.34). The first equivalence in (6.35) follows because the mean is then bounded above and away from zero; the second follows from (6.3). □

Warning 6.7 (The endpoint hypothesis already has RH strength). Because $\sqrt{\log N} = O_\varepsilon(N^\varepsilon)$, condition (6.33) itself implies $M(N) = O_\varepsilon(N^{1/2+\varepsilon})$ for every $\varepsilon > 0$, and hence implies RH. Conversely, the standard consequence of RH does not imply (6.33). Proposition 6.6 is therefore not a route from the original ratio ρ_Q to RH; it relates that ratio to weak Mertens only inside a pointwise regime already stronger than RH. The unconditional equivalence in Theorem 6.2 avoids this circularity by retaining the endpoint term in $\mathcal{B}_N$.

Remark 6.8 (The status of an asymptotic five-percent claim). A statement such as

$$\rho_Q(N) \le 0.05 + o(1) \tag{6.36}$$

would be a quantitative strengthening of bounded logarithmic second moment, not an elementary consequence of the finite model. Even the weaker assertion $\sup_N \rho_Q(N) < \infty$ requires, under (6.33), the weak Mertens conjecture. The finite data do not prove (6.36); they only show ratios near five percent over the stated cutoffs.

6.4. **Conditional spectral constant.** Under RH together with the negative-moment hypothesis $J_{-1}(T) \ll T$, Ng proves the logarithmic second-moment limit [21]

$$\frac{1}{Y}\int_0^Y e^{-y} M(e^y)^2\, dy \longrightarrow \beta_M, \tag{6.37}$$

$$\beta_M = 2\sum_{\gamma>0} \frac{1}{|\rho\zeta'(\rho)|^2}. \tag{6.38}$$

The passage from Ng's integral to the discrete normalization used here is explicit. Since $M(x) = M(n)$ on $n \le x < n+1$,

$$\sum_{n\le N} \frac{M(n)^2}{n^2} - \int_1^{N+1} \frac{M(x)^2}{x^2}\, dx = \sum_{n\le N} \frac{M(n)^2}{n^2(n+1)}. \tag{6.39}$$

The hypotheses of (6.37) include RH, and hence $M(n) \ll_\varepsilon n^{1/2+\varepsilon}$ for every $\varepsilon > 0$. Choosing $0 < \varepsilon < 1/2$ shows directly that the right side of (6.39) is dominated by the convergent series $\sum_n n^{-2+2\varepsilon}$. Hence the integral and discrete logarithmic means have the same conditional limit β_M.

A separate reported computation gives

$$2\sum_{k=1}^{400} \frac{1}{|\rho_k\zeta'(\rho_k)|^2} = 0.0287674, \tag{6.40}$$

$$\widehat{R}_{400} = \frac{2\widehat{c}}{T_{400}} = 0.0002715, \qquad \widehat{c} = 0.0923, \quad T_{400} = 679.74, \tag{6.41}$$

$$\widehat{\beta}_M = 0.0290389. \tag{6.42}$$

The second term is a tail ansatz, not a computed sum. It estimates the positive-ordinate density of $|\zeta'(\rho)|^{-2}$ from the last quarter of the 400 ordinates.

Gonek's conjecture is

$$J_{-1}(T) := \sum_{0<\gamma\le T} \frac{1}{|\zeta'(\rho)|^2} \sim \frac{3}{\pi^3} T, \tag{6.43}$$

where $3/\pi^3$ is the constant for positive ordinates [9, 21]. Substitution into the same tail ansatz gives

$$\frac{2(3/\pi^3)}{T_{400}} = 0.0002847, \qquad 0.0287674 + \frac{2(3/\pi^3)}{T_{400}} = 0.0290521. \tag{6.44}$$

The empirical $\widehat{c} = 0.0923$ is 4.6 percent below $3/\pi^3 = 0.0967546\ldots$, so there is no factor-two discrepancy.

Under the hypotheses needed for (6.37), together with $M(N)^2/[N\log N] \to 0$, the discrete–continuous identity (6.5) gives

$$\frac{\mathcal{B}_N(\mu)}{\log N} \longrightarrow \beta_M. \tag{6.45}$$

Combining this with Theorem 6.1 gives the orbit-normalized limit

$$\mathscr{R}_1(N) \longrightarrow \zeta(2)\beta_M. \tag{6.46}$$

Under the corresponding null-mean hypothesis, the original quadratic ratio has the same conditional limit:

$$\rho_Q(N) \longrightarrow \rho_\infty = \zeta(2)\beta_M. \tag{6.47}$$

The tail-corrected 400-zero value gives

$$\widehat{\rho}_\infty = 0.0477671, \qquad I(\widehat{\rho}_\infty) = 2.37284. \tag{6.48}$$

This explains why a ratio near five percent is plausible. It also answers the novelty question sharply: the approximate asymptotic percentage is a direct conditional combination of a classical spectral constant with the fixed-class mean d; the finite exact denominator and its comparison with the observed word are the model-specific contribution.

The reported spectral mass is concentrated at low ordinates:

$$\text{first zero: } 54.7\%, \qquad \text{first five: } 80.3\%, \qquad \text{first one hundred: } 97.4\%. \tag{6.49}$$

Thus the small quadratic scale is principally a low-frequency property. The same separate computation reports a zero-parameter reconstruction of the fourteen truncated values with RMS error $1.5\cdot 10^{-6}$. These spectral numbers require the separate reproducibility package specified in Appendix B; they are not inputs to Proposition 5.1.

Warning 6.9 (Quadratic mass does not control remote maxima). The constant in (6.38) is an ℓ^2-type norm of zero coefficients. It does not bound their ℓ^1-norm, force phase alignment, or control $\sup_x |M(x)|/\sqrt{x}$. Finite flatness of $\mathcal{Q}_{1000,N}$ therefore gives no evidence that the normalized Mertens maximum remains globally bounded.

7. Structural experiments beyond quadratic mass

The tests in this section ask whether the observed order contains information not already summarized by the quadratic functional. Their status is experimental, and their numerical specifications are recorded in Appendices A and B.

7.1. **Excursion orientation and shape.** For each completed excursion, let ε_r be its orientation, ℓ_r its support-time length, and

$$s_r = \sum_{n \text{ in excursion } r} \frac{1}{n} \tag{7.1}$$

its logarithmic integer duration. The tested orientation statistics were

$$\Theta = \frac{\sum_r s_r\varepsilon_r}{\sum_r s_r}, \tag{7.2}$$

$$P_1 = \frac{1}{K-1}\sum_{r<K} \varepsilon_r\varepsilon_{r+1}, \tag{7.3}$$

$$A_\ell = \mathrm{Corr}(\varepsilon_r, \log \ell_r). \tag{7.4}$$

Conditional randomization with the unsigned excursion skeleton fixed gave

$$p_\Theta = 0.574, \qquad p_{P_1} = 0.271, \qquad p_{A_\ell} = 0.479. \tag{7.5}$$

No orientation bias, nearest-neighbour persistence, or sign–length association was detected.

For unsigned excursion shapes, the tested coordinates were

$$A_r^{\mathrm{sh}} = \frac{1}{\ell_r^2} \sum_t H_r(t)^2, \tag{7.6}$$

$$H_r^{\max} = \frac{\max_t H_r(t)}{\sqrt{\ell_r}}, \tag{7.7}$$

$$\mathrm{HS}_r = \text{Horton–Strahler order}. \tag{7.8}$$

Five hundred observed excursions and 100 conditioned Catalan replicates gave

$$p_A = 0.505, \qquad p_H = 0.743, \qquad p_{\mathrm{HS}} = 0.485. \tag{7.9}$$

The tested shape statistics therefore do not distinguish the observed sample from the conditioned Catalan reference. Catalan enumeration and uniform Dyck paths are standard; see [25, 8].

7.2. **Scale partition and slow-envelope controls.** The partition audit in Table 6 compares an unweighted bin standard deviation with a Pearson-normalized statistic. The former has a two-sided value $p = 0.0066$ at 24 bins but not at 12 or 48 bins; the latter is nonsignificant at all three resolutions. There is therefore no partition-robust result.

For sliding windows, the short-scale statistics at window sizes 256 and 512 have correlations

$$r_{256} = 0.931, \qquad r_{512} = 0.914, \qquad p < 10^{-4} \quad \text{in both cases} \tag{7.10}$$

with the local trend $M(x)/\sqrt{x}$. At window size 1024, the reported split-half variance ratios are

$$(0.248,\ p = 0.111) \quad \text{and} \quad (0.706,\ p = 0.687). \tag{7.11}$$

The familywise result over seven scales from 256 through 2978 is

$$p_{\mathrm{global}} = 0.0645. \tag{7.12}$$

This familywise value is calibrated by recomputing the selected extremum over all seven scales inside each fixed-mask null realization. It does not treat the overlapping windows or scales as independent. Any pointwise window p-value without that replicate-level selection step would be invalid for the familywise question. These controls identify the smaller-scale feature with the known slow Mertens envelope and show that the larger-scale feature does not replicate between halves.

7.3. **Conditional-choice analysis.** For blocks of q nonzero entries, the feature set included second- and third-order Rényi entropies, zero local time, self-intersection count, path width, turn rate, the quadratic functional, transposition stability, and their prescribed interactions. Given a choice set $\mathcal{S}_r$, the conditional-logit model was

$$\mathbb{P}(\pi_j \mid \mathcal{S}_r) = \frac{\exp(\beta^{\mathsf{T}} X_{rj})}{\sum_\ell \exp(\beta^{\mathsf{T}} X_{r\ell})}. \tag{7.13}$$

See [18] for conditional logit and [2] for false-discovery control.

At $q = 256, 512, 1024$, the quadratic-only deviance tests gave

$$p_{\mathrm{ANOVA}} = (0.550,\ 0.904,\ 0.661), \tag{7.14}$$

with held-out gains per set

$$(1.0 \cdot 10^{-6},\ -4.16 \cdot 10^{-4},\ 0.00245). \tag{7.15}$$

The complete feature hierarchy did not produce a robust held-out selector. These results are distinct from the blind spectral experiment below. The choice sets overlap across blocks and scales; absent a component-level sandwich estimate or block bootstrap, the displayed deviance p-values are exploratory rather than confirmatory.

7.4. **Blind recovery of logarithmic frequencies.** Define the log-time signal

$$Y_\theta(t) = e^{-\theta t} M(\lfloor e^t \rfloor). \tag{7.16}$$

A hierarchical spectral procedure chose θ and frequencies from the Mertens path without reading zeta zeros or their derivatives. The external reference audit was performed only after fitting. Before the permutation null filter, the procedure matched 10 of the first 14 reference ordinates. For the first four it reported

reference ordinate	14.1347	21.0220	25.0109	30.4249
blind estimate	14.138	21.017	25.011	30.430

. (7.17)

The full fourteen-mode audit appears in Table 8.

The complete discovery, validation, and test pipeline was then rerun on 100 uniform fixed-mask permutations. A mode-by-mode filter at $q = 0.010$ retained

$$0 \text{ of the } 11 \text{ pre-null candidate modes.} \tag{7.18}$$

With the add-one empirical convention, however, the smallest attainable modewise value and the first Benjamini–Hochberg threshold were

$$\widehat{p}_{\min} = \frac{1}{100+1} = 0.00990099\ldots, \qquad \frac{q}{11} = 0.00090909\ldots. \tag{7.19}$$

Thus the design had no resolution for a sparse rank-one discovery at the declared q. At least

$$R \geq \left\lceil \frac{11}{0.010} \right\rceil - 1 = 1099 \tag{7.20}$$

complete placebo reruns are required merely to make the first BH threshold attainable. This count addresses resolution only. Because the modes are dependent and adaptively selected by a common pipeline, the standard independent/positive-dependence BH guarantee is not automatic. A prespecified global maximum statistic or a dependence-safe full-pipeline procedure is required for confirmatory inference.

Remark 7.1 (Interpretation of the blind spectrum). The reported pre-null frequency recovery should not be omitted, but it is not a discovery of previously unknown zeros. Classical explicit formulae place zero ordinates in the log-frequency content of $M(x)/\sqrt{x}$; a successful blind fit is therefore an algorithmic recovery of known spectral structure. The observation that no candidate survived the 100-placebo modewise filter is inconclusive, because (7.19) makes that filter underpowered by construction for a sparse discovery. The run neither establishes a new fixed-class spectral effect nor supplies strong evidence that no such effect exists. It is also not the ANOVA in (7.14): the two procedures answer different questions.

8. A route toward genuine arithmetic selection

8.1. **The exact selector already known.** The fixed-mask class discards the arithmetic relation that uniquely defines the Möbius function. That relation can be restored explicitly. For an arbitrary word w, define the augmented divisor-convolution defect

$$\Delta_N(w) \ = \omega_1 |w(1) - 1|^2 + \sum_{n=2}^{N} \omega_n \left| \sum_{d \mid n} w(d) \right|^2, \qquad \omega_n > 0. \tag{8.1}$$

Proposition 8.1 (Unique finite arithmetic selector)**.** *For every $N \geq 2$,*

$$\Delta_N(w) = 0 \quad \Longleftrightarrow \quad w(n) = \mu(n) \text{ for all } n \leq N. \tag{8.2}$$

Proof. The left side is equivalent to $w(1) = 1$ and $\sum_{d|n} w(d) = 0$ for every $2 \le n \le N$. These are the finite equations $(w * \mathbf{1})(n) = \delta_{n1}$. The Dirichlet inverse of the constant-one function is unique and equals μ. Equivalently, induction uses

$$w(n) = -\sum_{\substack{d|n \\ d<n}} w(d) \tag{8.3}$$

to determine each value. □

This proposition gives a unique position to M in the strongest possible sense, but it merely restates Möbius inversion. It contains no cancellation estimate and therefore supplies no progress toward RH.

8.2. **Arithmetic orientation calculus on the reflection fiber.** The defect Δ_N becomes nontrivial when it is used to resolve the orientation ambiguity left by the quadratic profile. For the r-th completed excursion, let

$$E_r = \{a_{\tau_{r-1}+1}, \dots, a_{\tau_r}\} \subseteq A_N, \qquad 1 \le r \le K, \tag{8.4}$$

and define its divisor-incidence signal by

$$g_r(n) = \begin{cases} \mu(1)\mathbf{1}_{\{1 \in E_r\}}, & n = 1, \\ \displaystyle\sum_{\substack{d|n \\ d \in E_r}} \mu(d), & 2 \le n \le N. \end{cases} \tag{8.5}$$

For $\eta = (\eta_1, \dots, \eta_K) \in \{0,1\}^K$, let w_η be obtained from μ by reflecting exactly those completed excursions for which $\eta_r = 1$. Every w_η lies in $\mathcal{O}_N(\mu)$, and hence has the same quadratic profile as μ.

Theorem 8.2 (Divisor-defect Gram form for excursion orientations)**.** *Define the $K \times K$ matrix*

$$(G_N)_{rs} = \sum_{n=1}^{N} \omega_n g_r(n) g_s(n). \tag{8.6}$$

Then G_N is positive semidefinite and

$$\Delta_N(w_\eta) = 4\,\eta^{\mathsf{T}} G_N \eta \qquad (\eta \in \{0,1\}^K). \tag{8.7}$$

Moreover,

$$\eta^{\mathsf{T}} G_N \eta > 0 \qquad \text{for every } \eta \neq 0. \tag{8.8}$$

Proof. For $n \ge 2$, Möbius inversion gives $\sum_{d|n} \mu(d) = 0$. Reflecting the selected excursion blocks changes the divisor sum to

$$\sum_{d|n} w_\eta(d) = -2 \sum_{r=1}^{K} \eta_r g_r(n). \tag{8.9}$$

The same identity at $n = 1$ holds for the augmented term $w_\eta(1) - 1$. Substitution into (8.1) gives (8.7). Equation (8.6) is a weighted Gram matrix, so it is positive semidefinite. If a nonzero Boolean vector had zero quadratic form, then $\Delta_N(w_\eta) = 0$; Proposition 8.1 would force $w_\eta = \mu$, contrary to the fact that a nonempty set of completed excursions was reflected. □

The diagonal entries of G_N measure the individual arithmetic cost of reflecting one excursion. Its off-diagonal entries measure coupling between two excursion orientations through common multiples. Thus the two exact local calculi in the paper address complementary layers:

$$\begin{aligned} \text{opposite-sign transposition} &\longmapsto \text{change of quadratic path mass,} \\ \text{excursion reflection} &\longmapsto \text{change of divisor-convolution defect.} \end{aligned} \tag{8.10}$$

The second formula does not prove cancellation, because it starts from the exact Möbius identity. Its value is structural: on a fiber where every even path statistic is constant, it identifies the arithmetic orientation problem with a finite positive-semidefinite Boolean quadratic form.

8.3. **Projective uniqueness is not a growth estimate.** RH is a statement about one projectively consistent infinite arithmetic sequence. A member of $\Omega_N^{(1)}$ is only a finite word, and the uniform classes for successive N do not define a canonical infinite extension. It is therefore not mathematically meaningful to partition one finite class into words that satisfy or violate RH without first prescribing an extension law.

There is nevertheless an exact projective uniqueness statement. Define

$$\mathscr{P} = \left\{ w \in \{-1,0,1\}^{\mathbb{N}} : w|_{[1,N]} \in \Omega_N^{(1)} \text{ for every } N \right\}. \tag{8.11}$$

Proposition 8.3 (Projective intersection of the fixed-count classes)**.**

$$\mathscr{P} = \{\mu\}. \tag{8.12}$$

Proof. Membership at every cutoff fixes $R_w(N) = M(N)$ for every N. Therefore

$$w(N) = R_w(N) - R_w(N-1) = M(N) - M(N-1) = \mu(N). \tag{8.13}$$

□

This uniqueness does not imply RH. Let

$$\mathscr{A} = \{w : \kappa_0(w) = 0\}, \qquad \mathscr{C} = \{w : \Delta_N(w) = 0 \text{ for every } N\}. \tag{8.14}$$

Möbius inversion says $\mathscr{C} = \{\mu\}$. RH says

$$\mu \in \mathscr{A}, \tag{8.15}$$

or equivalently $\mathscr{C} \subseteq \mathscr{A}$. Knowing that $\mathscr{C}$ is a singleton supplies uniqueness but not the membership (8.15). Conversely, the literal assertion $\mathscr{A} = \{\mu\}$ is false: many nonarithmetic deterministic and random words have square-root-scale partial sums.

For an infinite sequence w, one may define its growth exponent

$$h(w) = \limsup_{x\to\infty} \frac{\log(1 + |R_w(x)|)}{\log x}. \tag{8.16}$$

For the Möbius function, RH is equivalent to $h(\mu) \le 1/2$. The stronger weak-Mertens property is

$$\sup_{N\ge 2} \frac{1}{\log N} \sum_{n\le N} \frac{R_w(n)^2}{n^2} < \infty. \tag{8.17}$$

For arbitrary paths, (8.17) does not control the supremum because sparse spikes can be hidden in an average. Its implication for RH is specifically arithmetic and uses the Mellin transform of M and $1/\zeta$.

8.4. **A hierarchy of conditioning levels.** For a prescribed set of cutoffs $\mathcal{C} \subseteq [1,N]$ and tolerance $\delta \ge 0$, define

$$\mathcal{P}_N(\mathcal{C},\delta) = \left\{ w \in \Omega_N^{(1)} : \max_{x\in\mathcal{C}} |\mathscr{A}_x(w) - \mathscr{A}_x(\mu)| \le \delta \right\}. \tag{8.18}$$

The model therefore leads to the exact finite hierarchy

$$\Omega_N^{(1)} \supseteq \mathcal{P}_N(\mathcal{C},\delta) \supseteq \mathcal{P}_N([1,N],0) = \mathcal{O}_N(\mu) \supseteq \{\mu\}. \tag{8.19}$$

The second inclusion holds whenever the observed exact profile is admitted, as it is for every $\delta \ge 0$. The first level fixes the mask and sign counts. The full, exact profile fixes the absolute path; a coarser profile leaves a larger tolerance class. The last step requires reflection-sensitive arithmetic information. Theorem 8.2 describes that final step exactly for the complete divisor defect, while

the nontrivial research question is whether a strictly local or low-complexity portion of the divisor relations already distinguishes the observed orientations.

8.5. **Exact divisor-residual stability and the missing estimate.** The complete divisor defect yields an exact quantitative identity, not merely a binary selector. Put

$$r_w(n) = \sum_{d|n} w(d) - \delta_{n1}. \tag{8.20}$$

With the convention in (8.1),

$$\Delta_N(w) = \sum_{n\le N} \omega_n r_w(n)^2. \tag{8.21}$$

Proposition 8.4 (Divisor residual to path deviation)**.** *For every word w defined on the positive integers,*

$$w - \mu = \mu * r_w, \tag{8.22}$$

$$R_w(x) - M(x) = \sum_{d\le x} r_w(d) M\left(\left\lfloor \frac{x}{d} \right\rfloor\right). \tag{8.23}$$

Consequently, for arbitrary prescribed weights $\omega_d > 0$,

$$|R_w(x) - M(x)|^2 \le \left(\sum_{d\le x} \omega_d r_w(d)^2\right)\left(\sum_{d\le x} \frac{M(\lfloor x/d\rfloor)^2}{\omega_d}\right). \tag{8.24}$$

The same statements hold for a word given only through N, provided $x \le N$.

Proof. Since $\mathbf{1} * \mu = \delta$,

$$r_w = \mathbf{1} * w - \delta = \mathbf{1} * (w - \mu). \tag{8.25}$$

Convolution with μ proves (8.22). Summing that identity through x and reversing the finite divisor sum gives (8.23). Equation (8.24) is weighted Cauchy–Schwarz. □

Proposition 8.4 is falsifiable and quantitative: it measures how an approximate divisor inverse can deviate from the Mertens path. It does not imply RH. Its amplification factor still contains the unknown values of M, and at $w = \mu$ both sides of (8.23) vanish identically.

A genuinely coercive theorem would have to fix its form in advance. For example, after prescribing ω_n, constants α, θ, and a domain of words independently of μ, one could ask whether

$$\mathcal{B}_N(w) \le C_\varepsilon N^\varepsilon + C N^\alpha \Delta_N(w)^\theta \tag{8.26}$$

holds uniformly. Applied at $\Delta_N(\mu) = 0$, a bound of this strength would contain the RH estimate itself, so it cannot be treated as a routine lemma. Unlike an inequality with an unspecified remainder function, (8.26) is at least a definite statement that can be proved or disproved.

A finite intermediate problem is obtained from the relaxed classes

$$\Omega_N(\eta) = \{w \in \{-1, 0, 1\}^{[1,N]} : \Delta_N(w) \le \eta\} \tag{8.27}$$

by studying the two extremal functions

$$\inf_{w\in\Omega_N(\eta)} \mathcal{B}_N(w), \qquad \sup_{w\in\Omega_N(\eta)} \mathcal{B}_N(w), \tag{8.28}$$

or their analogues for $\mathcal{Q}$. Any nontrivial uniform relation between these quantities would connect multiplicative compatibility to path geometry without merely restating exact inversion.

A statistically meaningful intermediate programme has three parts:

(i) condition simultaneously on the mask, endpoint, and quadratic-mass profile over many nested cutoffs, rather than on one scalar value;

(ii) test coordinates orthogonal to the absolute path, such as orientation bits and cross-level nesting, under exact conditional nulls;
(iii) require projective consistency in N and a deterministic link from any surviving coordinate to the divisor-convolution defect or to (6.10).

The orientation and shape tests in Section 7 do not detect a departure on the available interval. The blind spectrum recovers known analytic structure but does not provide an independent combinatorial selector. Thus no current statistic uniquely distinguishes the Möbius path from all RH-compatible paths, let alone from paths satisfying a stronger mean-square property.

Question 8.5 (Projectively consistent residual selector)**.** Does there exist a statistic $\mathcal{S}_N(w)$, not determined by the mask, endpoint, or multiscale quadratic-mass profile, whose conditional null is explicit, whose value is consistent under increasing cutoffs, and for which a deterministic inequality links $\mathcal{S}_N(\mu)$ to (6.12)?

Question 8.6 (Low-complexity orientation selector)**.** Does a prescribed local or truncated subset of the divisor–multiple relations produce a reflection-sensitive functional that uniquely selects the observed orientation inside $\mathcal{O}_N(\mu)$, with a complexity substantially smaller than the complete inversion system?

9. Conclusion

The contribution is not merely the evaluation of a statistic at a large cutoff. The fixed-mask, fixed-count orbit supports an exact finite theory: the mean and variance of the logarithmic quadratic path mass, the factored and semiseparable squared-prefix covariance, a one-pass variance formula, the order-dependent bilinear decomposition, the transposition formula, the closed-form global minimum, M-convex local–global optimality, the absolute-path quotient, the excursion-reflection fiber, exact return counts, and the divisor-defect Gram form for orientation bits. These statements hold for every finite sign population satisfying their stated hypotheses.

There are also two infinite statements. First, proportional windows have a rigorous weighted Brownian-bridge quadratic limit with the explicit spectrum in (3.59). This is an asymptotic theorem about the permutation family, although it does not cover the fixed-N_0 lower-tail regime. Second, the exact orbit scales are asymptotic to $d \log N$, and

$$\mathrm{RH} \quad \Longleftrightarrow \quad \kappa_\Omega(\mu) = 0. \tag{9.1}$$

Equation (9.1) is a model-specific normalization of the classical Mertens criterion, not a new estimate toward RH. It separates RH-incompatible polynomial growth, RH-scale subpolynomial growth, and the stronger bounded-ratio regime. Many other words have exponent zero; the equivalence acquires number-theoretic content only at the distinguished word μ.

On the identified data, the Möbius word has 5.18 percent of its exact class mean at the largest cutoff, with the sign of the finite separation robust under the declared forward-error audit. For an expert in analytic number theory, the magnitude near five percent is not a new asymptotic constant: under Ng's hypotheses it is explained by $\zeta(2)\beta_M$, and the first few zeros dominate β_M. The model-specific part is the exact conditional denominator and the geometry of the orbit. The sampled lower tail is much thinner than the rejected two-moment proxy, but its exact finite probability remains unknown.

Projective membership in every observed fixed-count class already forces $w = \mu$, and the complete divisor defect also selects μ uniquely. Neither uniqueness proves RH. In the notation of (8.14), inversion proves $\mathscr{C} = \{\mu\}$, whereas RH asks for the separate membership $\mathscr{C} \subseteq \mathscr{A}$. Proposition 8.4 gives an exact stability operator between the two structures, but its amplification factor still contains M. Replacing that factor by an a-priori coercive estimate of RH strength is the unresolved analytic step.

No tested excursion orientation, Catalan shape, scale partition, or local conditional-choice statistic supplies such an estimate. The blind spectral procedure recovers several known zero frequencies without using them in the fit, but its permutation-corrected mode filter retains none; with only 100 placebos, that filter is too coarse for a sparse BH discovery and the outcome is inconclusive. Thus the paper provides exact orbit geometry, algorithms, and two asymptotic formulations, while making no claim to prove RH, weak Mertens, or an unconditional five-percent limit.

Appendix A. Numerical tables

This appendix collects the numerical outputs used in the text. Exact finite-class formulae are distinguished from sampled statistics in every caption.

Table 1. Observed quadratic path mass and exact fixed-mask, fixed-count class moments, with $N_0 = 1000$. Decimal entries are binary64 evaluations of the exact formulae.

N	$M(N)$	$\mathcal{Q}(\mu)$	$\overline{\mathcal{Q}}_N$	σ_N	ρ_Q	Z_Q
10,000	−23	0.032174	0.39129	0.37891	0.082224	−0.9478
30,000	18	0.030803	0.43847	0.39324	0.070251	−1.0367
100,000	−48	0.031084	0.48239	0.39632	0.064437	−1.1387
300,000	220	0.029369	0.53006	0.39912	0.055407	−1.2545
1,000,000	212	0.030100	0.52665	0.37332	0.057154	−1.3301
5,000,000	−709	0.029515	0.54848	0.35510	0.053812	−1.4614
50,000,000	−908	0.029733	0.55336	0.32471	0.053732	−1.6126
100,000,000	1,928	0.029641	0.55844	0.31817	0.053079	−1.6620
150,000,000	1,089	0.029536	0.55767	0.31333	0.052963	−1.6855
200,000,000	320	0.029772	0.55824	0.31038	0.053332	−1.7027
250,000,000	−982	0.029944	0.55941	0.30838	0.053529	−1.7169
300,000,000	−592	0.029662	0.55989	0.30663	0.052977	−1.7292
350,000,000	2,896	0.029508	0.56226	0.30578	0.052482	−1.7423
455,052,511	−5,414	0.029334	0.56628	0.30436	0.051800	−1.7642

A.1. **Fixed-class quadratic comparison.** The complete-cutoff support contains 276,638,778 nonzero entries. The corresponding squarefree density is 0.60793 to the displayed precision. All fourteen parity checks

$$M(N) \equiv Q(N) \pmod 2 \tag{A.1}$$

passed.

Table 2. Uniform fixed-count sampling. Positive-hit intervals are Wilson intervals [27]; zero-hit entries are exact one-sided 95-percent Clopper–Pearson bounds.

N	draws	hits	$\widehat{P}_Q$	95-percent interval or bound	Z_s
10,000	300,000	412	0.0013733	[0.0012470,0.0015124]	1.3216
30,000	150,000	15	0.0001000	[0.0000606,0.0001650]	0.9930
100,000	100,000	0	0	$< 2.9957 \cdot 10^{-5}$	−1.6256
300,000	50,000	0	0	$< 5.9913 \cdot 10^{-5}$	−0.7217
1,000,000	10,000	0	0	$< 2.9953 \cdot 10^{-4}$	−1.9583
5,000,000	10,000	0	0	$< 2.9953 \cdot 10^{-4}$	−1.2327

A.2. **Uniform-permutation ladder.** Here $Z_s = (s - \sigma_N)/\operatorname{SE}(s)$, with $\operatorname{SE}(s)$ estimated from the sample fourth moment. The first 8,200 draws at $N = 5 \cdot 10^6$ have

$$\overline{\mathcal{Q}}_{\text{sample}} = 0.544194764161, \qquad s_{\text{sample}} = 0.350499677391. \tag{A.2}$$

These are compatible with the exact mean 0.548481916755 and exact standard deviation 0.35510. The value 0.350195 is the leading $L_N - 5/2$ asymptotic standard deviation, not the exact finite-class value.

TABLE 3. Recorded scale coefficients. The second and third comparison columns arise from the feedback model and its scale adjustment; they are not independent estimates of $\zeta(2)\mathcal{Q}$.

K	c_{observed}	$c_{a=4.8161}$	$c_{a,\text{adjusted}}$
256	0.048729	0.047858	0.064982
512	0.067637	0.054504	0.066281
1024	0.036207	0.048046	0.065380
2048	0.049998	0.053648	0.037300
4096	0.054833	0.043456	0.050333
8192	0.044089	0.045434	0.046309
16,384	0.049748	0.050261	0.040051
32,768	0.038838	0.043234	0.044990
65,536	0.061641	0.050747	0.034831
131,072	0.039820	0.047357	0.062011
262,144	0.036670	0.047253	0.036656
524,288	0.058102	0.046882	0.035871

A.3. **Support-time coefficients.** The chosen reference

$$c_0 = \frac{0.0300}{6/\pi^2} = \frac{\pi^2}{200} = 0.049348022005\ldots \tag{A.3}$$

was fixed by the rounded target 0.0300, not estimated independently. The first fitted log-scale mode was

$$g_1(t) = -0.000774584 - 0.009187853\cos(2.541179t) - 0.006146902\sin(2.541179t), \qquad t = \log_2 K. \tag{A.4}$$

Its period is 2.47255 octaves, amplitude 0.011054, and in-sample $R^2 = 0.643529$. The complete causal forecast did not consistently improve on a constant predictor.

The separate spectral coefficient estimates are

K	direct	cross-fitted
14	0.11802	0.11793
50	0.10508	0.10508
100	0.10250	0.10270
102	0.10211	0.10231

. (A.5)

No equality between these spectral coefficients and (3.67) is asserted.

A.4. **Excursion diagnostics.** At $N = 5 \cdot 10^6$,

$$m = 3{,}039{,}633, \qquad p = 1{,}519{,}462, \qquad q = 1{,}520{,}171. \tag{A.6}$$

There are 6,016 completed excursions in the full support-time path, so

$$\log_{10}|\mathcal{O}_N(\mu)| = 6016\log_{10} 2 = 1810.996\ldots. \tag{A.7}$$

A windowed analysis excluding excursions meeting $n \le 1000$ contains 5,956 completed excursions.

A reported sample of 300 uniform bridges gave

$$\overline{Z} = 1679.8, \qquad s_Z = 1102.4, \qquad \min Z = 7, \qquad \max Z = 5544, \tag{A.8}$$

so none reached the observed value 6,016. The corresponding exact one-sided 95-percent zero-exceedance bound is

$$1 - 0.05^{1/300} = 0.00994. \tag{A.9}$$

For the longest completed-excursion fraction, the same sample had mean 0.5301, median 0.4806, and minimum 0.1219, compared with the observed 0.11737.

TABLE 4. Observed excursion-length tails and the unconditioned first-return benchmark $6016\sqrt{2/\pi}\,c^{-1/2}$.

threshold c	observed	benchmark	ratio
10	1474	1517.9	0.971
100	454	480.0	0.946
1000	140	151.8	0.922
10,000	40	48.0	0.833

TABLE 5. Large-cutoff support and return census. The complete-cutoff return count was not recorded in the cited census.

N	nonzero entries	returns to zero	$M(N)$
50,000,000	30,396,344	15,756	−908
100,000,000	60,792,694	25,525	1,928
150,000,000	91,189,087	34,232	1,089
200,000,000	121,585,426	36,069	320
250,000,000	151,981,736	36,755	−982
300,000,000	182,378,126	42,505	−592
350,000,000	212,774,472	47,215	2,896
455,052,511	276,638,778	not recorded	$-5,414$

Large computations of the Mertens function provide numerical context for these finite ranges [5, 16, 15].

TABLE 6. Partition audit. Each p-value is two-sided; the Pearson statistic has approximately one fewer degree of freedom than the number of bins.

bins	unweighted: observed/null	p	Pearson: observed/null	p
12	0.252/0.352	0.135	6.07/11.06	0.188
24	0.351/0.512	0.0066	15.65/23.24	0.103
48	0.706/0.710	0.837	45.89/47.56	0.739

A.5. **Partition and conditional-choice diagnostics.**

A.6. **Blind frequency audit.**

APPENDIX B. REPRODUCIBILITY SPECIFICATION

This appendix specifies the complete deposit required to reproduce every computational statement in the paper. It is a protocol, not a claim that an independent rerun was performed while preparing this version. The exact finite identities can be checked without the data; the decimal tables require the identified inputs and the declared software environment.

TABLE 7. Held-out audit of the local quadratic-only conditional-choice model. Each scale used 1,000 choice sets with 13 paths.

q	deviance p	test gain per set	test top–1
256	0.550	$1.0 \cdot 10^{-6}$	0.120
512	0.904	$-4.16 \cdot 10^{-4}$	0.040
1024	0.661	0.00245	0.120

TABLE 8. External audit of the first fourteen zero ordinates against the pre-null blind spectral candidates. Reference values were not used during fitting. A dash denotes no matched candidate.

mode	reference	blind estimate	error
1	14.135	14.138	0.00347
2	21.022	21.017	−0.00461
3	25.011	25.011	0.00048
4	30.425	30.430	0.00471
5	32.935	32.931	−0.00422
6	37.586	37.559	−0.02737
7	40.919	–	–
8	43.327	–	–
9	48.005	48.015	0.00970
10	49.774	49.837	0.06298
11	52.970	–	–
12	56.446	–	–
13	59.347	59.377	0.03029
14	60.832	60.810	−0.02195

B.1. **Required archive structure.** The reproducibility archive shall contain the following top-level items:

(i) `README.md`, with the single-command instructions, expected wall time, memory requirement, and a map from each table and equation in the paper to its raw output;
(ii) `MANIFEST.sha256`, containing a SHA-256 digest for every input, source file, raw output, and generated table;
(iii) `environment.txt`, recording operating system, processor, available memory, MATLAB release and update, BLAS library, thread count, and locale;
(iv) `data/moebius.bin`, the signed-byte Möbius word;
(v) `data/zeta_zeros_400.csv` and `data/zeta_derivatives_400.csv`, used only by the separate spectral branch;
(vi) `src/`, containing the canonical source files listed below;
(vii) `raw/`, containing unmodified console logs, CSV files, and MAT files from every run;
(viii) `tables/`, containing only deterministic transformations of the raw files into manuscript tables;
(ix) `tests/`, containing the finite exhaustive tests and their expected outputs.

No source file may download data during a run. The exact/permutation branch and the spectral branch must have separate entry points, and the former must abort if a spectral input is opened.

B.2. **Canonical source inventory.** The archive shall use mathematical names and contain, at minimum,

`run_all_reproductions.m`:
Top-level orchestrator; runs tests, exact calculations, sampling experiments, and table generation in a fixed order.

`fixed_mask_moments.m`:
Streaming observed sum, exact class mean, exact variance, forward-error audit, continuum diagnostics, and all fourteen cutoffs.

`independent_mobius_certificate.c`:
Second implementation, sharing no sieve or streaming code with the MATLAB branch, which verifies every byte of `moebius.bin` in documented blocks and writes block hashes and mismatch counts.

`uniform_bridge_ladder.m`:
Uniform fixed-count sampler at the six Monte Carlo cutoffs.

`excursion_statistics.m`:
Completed-excursion decomposition, reflection count, exact return mean, orientation randomization, Catalan controls, and length diagnostics.

`partition_and_scale_tests.m`:
Audits with 12, 24, and 48 bins. Sliding scales, slow-envelope correlations, split halves, and familywise calibration.

`conditional_choice_analysis.m`:
Feature construction, conditional-logit models, chronological splits, and deviance tables.

`blind_log_spectrum.m`:
Blind discovery/validation/test spectral pipeline and the 100 complete fixed-mask placebo reruns.

`spectral_second_moment.m`:
Four-hundred-zero direct sum, tail alternatives, fourteen-cutoff reconstruction, and residual decomposition.

`make_manuscript_tables.m`:
Reads raw outputs and writes tables without recomputing any statistic.

Every function used by these entry points must be included as source rather than supplied through an unrecorded local path. The archive must also state the license under which the sources and derived tables are released.

B.3. **Input definition and integrity checks.** The main data file has the immutable specification

$$\begin{aligned} \text{format} &= \texttt{int8}, \\ \text{allowed values} &= \{-1, 0, 1\}, \\ \text{number of bytes} &= 455{,}052{,}511, \\ \text{SHA-256} &= \texttt{0e551f47d924ef6f3e348e131fdf53e0c677ad5e5471ef9d6249b3fd662e2ea4}. \end{aligned} \tag{B.1}$$

Before any statistic is evaluated, the program must:

(a) verify the byte count, allowed alphabet, and SHA-256 digest;
(b) independently sieve $\mu(n)$ for $1 \le n \le 10^6$ and compare every entry;
(c) verify every remaining entry with the independent full-file implementation, recording a digest and mismatch count for every block;
(d) verify at each cutoff the nonzero count and endpoint recorded in the raw reference table;
(e) abort unless
$$M(N) \equiv Q(N) \pmod 2 \tag{B.2}$$
at every cutoff;
(f) verify $p = (m + S)/2$, $q = (m - S)/2$, and $T_m = S$.

A SHA-256 digest identifies one byte string; it does not prove that the string is the Möbius sequence through the stated cutoff. The first-million sieve, cutoff counts, endpoints, and parity

tests are strong corruption checks but do not certify the untested tail. Until the full-file independent comparison and a public immutable deposit are available, every numerical statement in this paper is explicitly conditional on the identified file. The symbolic finite-orbit theorems are unaffected.

The spectral CSV files must contain one row per positive ordinate with columns `index`, `gamma`, `ReZetaPrime`, `ImZetaPrime`, `AbsZetaPrime`, working precision, software used, and an independent residual or interval check for $\zeta(1/2+i\gamma)$. Their hashes must be listed in `MANIFEST.sha256`. The code must recompute $|\zeta'(\rho)|$ from its real and imaginary parts and abort if it disagrees with the stored absolute value beyond the declared tolerance.

B.4. **Exact-layer configuration.** The main run uses

$$N_0 = 1000, \tag{B.3}$$

$$\begin{aligned}\mathcal{N}_{\mathrm{exact}} = \{&10^4, 3\cdot 10^4, 10^5, 3\cdot 10^5, 10^6, 5\cdot 10^6,\\ &5\cdot 10^7, 10^8, 1.5\cdot 10^8, 2\cdot 10^8, 2.5\cdot 10^8, 3\cdot 10^8, 3.5\cdot 10^8, 455052511\}.\end{aligned} \tag{B.4}$$

The streaming read block is 2^{22} bytes and the positive-sum audit block is 2^{16} terms. Products that can exceed 2^{53} must not be formed. In particular, the implementation shall use

$$\left(\frac{M(n)}{n}\right)^2, \qquad \frac{Q(n)}{n}\frac{Q(n)-1}{n}, \qquad \frac{Q(n)}{n}\frac{1}{n} \tag{B.5}$$

rather than forming $M(n)^2$, $Q(n)^2$, or n^2 first at the largest cutoffs.

The mean must be computed independently in two ways:

(a) from the streaming sums Σ_1, Σ_2 in (3.7)–(3.8);
(b) from the explicit plateau weights and (3.11).

The variance must likewise be evaluated by the prefix formula (3.13) and, at $N = 10^4$, by the literal triangular $O(m^2)$ sum (3.14).

B.5. **Floating-point audit.** The recorded main environment is MATLAB 24.2.0.2740171 (R2024b Update 1). The forward-error argument assumes IEEE-754 binary64, round-to-nearest ties-to-even, no extended precision, no fused-operation contraction, and no reassociation. If the execution environment does not satisfy these conditions, the program must label the audit inapplicable or replace it with directed-rounding interval arithmetic.

Within each block of r positive summands, ordinary summation is bounded using

$$\gamma_r = \frac{ru}{1-ru}, \qquad u = 2^{-53}, \tag{B.6}$$

and block totals are accumulated with compensated summation. Bounds are propagated through the algebra for $\overline{\mathcal{Q}}_N$ and $\Delta_N = \mathcal{Q}(\mu) - \overline{\mathcal{Q}}_N$. The run must abort unless

$$0 \notin [\Delta_N^-, \Delta_N^+] \quad \text{and} \quad \Delta_N^+ < 0 \tag{B.7}$$

at every cutoff. This procedure is an a-priori forward-error audit in the sense of [11]; it is not interval arithmetic with directed rounding.

The bound must include, separately, exact integer-to-binary64 conversion, rounding in every division and multiplication used to form a summand, rounding in block accumulation, compensated accumulation of block totals, normalization by L_N, and the final subtraction defining Δ_N. The γ_r term alone bounds only summation of already rounded inputs. Agreement with a compensated or differently ordered sum is a sensitivity cross-check and is not itself a rigorous error bound.

B.6. **Mandatory exact self-tests.** The test suite must run before the data file is processed and must abort on any failure. It includes:

(i) complete enumeration of every fixed-count sign class with $4 \le m \le 10$, covering 56 classes and 1,890 covariance pairs;
(ii) comparison of every enumerated value $\mathrm{Cov}(T_j^2, T_k^2)$ with (2.15); the recorded maximum absolute error is $3.2 \cdot 10^{-13}$;
(iii) explicit enumeration of the 20 words with $m = 6, p = q = 3$, including $\mu_2 = -1/5$, $\mu_4 = 1/5$, and

$$\mathrm{Var}(T_2^2) = 3.84; \tag{B.8}$$

(iv) checks that $\psi(m) = 0$ and $\mathbb{E}T_m^2 = S^2$ identically;
(v) equality of the $O(m)$ and $O(m^2)$ variance implementations at $N = 10^4$, with recorded relative discrepancy $3.05 \cdot 10^{-16}$;
(vi) equality of the streaming and plateau-weight mean and variance routes, with recorded mismatches between $7 \cdot 10^{-18}$ and 10^{-16};
(vii) exhaustive checks of Proposition 4.5 for at least $(p, q) = (3, 3), (4, 3), (5, 4), (6, 4)$;
(viii) a SHA-256 unit test against a published standard test vector;
(ix) a symbolic or high-precision numerical self-test of the Legendre transform in (5.19), explicitly labelled as a transform test rather than a proof of the OU transfer.

The reference arithmetic printed by the test suite is

$$1 - 0.05^{1/2800} = 0.00106933223978, \tag{B.9}$$

$$(1 - 0.00428)^{2800} = 6.08478495942 \cdot 10^{-6}, \tag{B.10}$$

$$1 - 0.05^{1/9500} = 0.000315290524814, \tag{B.11}$$

$$9500(0.00428) = 40.66. \tag{B.12}$$

B.7. **Uniform sampling protocol.** The Monte Carlo configuration is

N	10^4	$3 \cdot 10^4$	10^5	$3 \cdot 10^5$	10^6	$5 \cdot 10^6$
R	300000	150000	100000	50000	10000	10000
seed	19092021	19092022	19092023	19092024	19092025	19092026

. (B.13)

The generator is MT19937, invoked as `rng(seed,'twister')` separately at each cutoff. A draw is produced by a uniform random permutation of the p positive and q negative signs, followed by placement on the fixed support. The program must check $T_m = S$ on every draw and test the marginal positive-sign frequency at fixed support indices.

For a positive hit count, the archive reports the Wilson interval and the exact binomial likelihood. For zero hits, it reports the one-sided exact bound

$$p_{\mathrm{upper}} = 1 - \alpha^{1/R}, \qquad \alpha = 0.05. \tag{B.14}$$

The sample standard-deviation check is a plug-in delta-method diagnostic, not an exact Z-test. It uses

$$\mathrm{SE}(s) = \frac{s}{2}\sqrt{\frac{\widehat{\kappa} - 1}{R}}, \qquad \widehat{\kappa} = \frac{\widehat{m}_4}{s^4}. \tag{B.15}$$

The program must print the diagnostic $(s - \sigma_N)/\,\mathrm{SE}(s)$ and require its absolute value to be below the declared calibration threshold; any inferential use additionally requires a bootstrap or permutation calibration.

The two-moment comparison uses

$$\nu = \frac{2\overline{\mathcal{Q}}_N^2}{\sigma_N^2}, \qquad \mathcal{Q}_{\chi^2} = \frac{\overline{\mathcal{Q}}_N}{\nu}\chi_\nu^2. \tag{B.16}$$

It is a deliberately falsifiable approximation, not the sampling law. Every sampled quadratic value, not only summary statistics, must be written to a per-cutoff CSV file.

MT19937 has 19,937 state bits and cannot visit every member of the largest class. The archive shall state this limitation. The exact mean and variance are unaffected; the Monte Carlo results concern the implemented uniform finite-permutation algorithm and require an independent generator as a robustness check before publication.

B.8. **Excursion and conditional-choice protocols.** The excursion archive must include:

(a) the complete ordered list of return times and completed-excursion lengths;
(b) a flag showing whether the terminal segment is incomplete and excluded from reflection;
(c) 100,000 orientation-null realizations with the unsigned skeleton fixed;
(d) the 500 observed excursions and all 100 Catalan replicates used for the shape tests;
(e) all 300 uniform bridges used for the return-count and longest-excursion summaries;
(f) the exact seed and generator state for each component.

The archive must state the rule selecting the 500 observed excursions, stratify or condition on their lengths, and generate each reference path uniformly from the irreducible Dyck excursions of the matching length. Pooling across lengths without recording the mixture is not permitted.

The conditional-choice archive must include all 1,000 choice sets with 13 paths at each $q = 256, 512, 1024$, the feature matrix before and after standardization, the chronological split indices, coefficient estimates, standard errors, log likelihoods, deviance increments, degrees of freedom, held-out log scores, and false-discovery adjustment. Expected cell counts must be reported for every chi-square diagnostic; where they are too small, an exact or permutation calibration is required.

B.9. **Blind spectral protocol.** The blind log-spectrum run uses $N = 5 \cdot 10^6$, 1,321 logarithmic samples, frequency ceiling 216.275, four hierarchical levels, and growing capacity

$$J(n) = \lfloor n^{0.38} \rfloor, \qquad J_{\max} = 15. \tag{B.17}$$

The chronological discovery, validation, and test endpoints are $0.45N, 0.65N, N$. Reference ordinates must not be loaded until the blind fit and all 100 placebo fits are complete. Each placebo must rerun the entire selection pipeline, including exponent choice, peak selection, frequency refinement, and validation. The archive must preserve:

(a) all 1,321 sampled log-time values and observed signals;
(b) every candidate peak before and after hierarchical selection;
(c) all 100 placebo candidate sets and test gains;
(d) the mode-by-mode multiple-testing table;
(e) the external reference audit generated only after selection;
(f) warnings, ranks, and condition numbers from every linear solve.

The pre-null match table is an algorithmic recovery result. The recorded 100-placebo run is a pilot resolution audit only: by (7.19), zero surviving modes cannot be used as confirmatory evidence of absence. At least 1,099 complete placebo reruns are needed merely to resolve the first BH threshold at $q = 0.010$; more are needed for stable tail estimation, and that increase alone does not resolve adaptive dependence. A confirmatory analysis must either justify a dependence-safe modewise procedure or prespecify one global maximum statistic and calibrate that single statistic with the full-pipeline placebo distribution. The design must be fixed before a new run and recorded in the manifest.

B.10. **Separate spectral-second-moment protocol.** The spectral run must use arbitrary precision high enough that rounding in each term of (6.40) is negligible at the printed precision. It must output:

(a) all 400 individual values $2/|\rho_k \zeta'(\rho_k)|^2$ and their cumulative sums;
(b) the contributions of the first, first five, and first one hundred zeros;

(c) the precise last-quarter estimator defining $\widehat{c} = 0.0923$;
(d) both tail alternatives $\widehat{c}$ and $3/\pi^3$;
(e) the fourteen reconstructed quadratic-mass values, their residuals, and the zero-parameter RMS;
(f) the fit $R_N = A + B/L_N$, its covariance matrix, and the discrete sum–integral correction;
(g) all working-precision and zero-validation diagnostics.

No output from this branch may be merged with the no-zero exact run before both raw manifests have been written.

B.11. **Raw output inventory and acceptance criteria.** At minimum, the archive shall preserve

```
exact_layer.csv, permutation_ladder.csv, sampled_quadratic_values.csv,
continuum_diagnostics.csv, ou_quantiles.csv, excursions.csv,
partition_audit.csv, conditional_choice.csv, blind_modes.csv,
blind_placebos.csv, spectral_terms.csv.
```
(B.18)

The MAT files must retain every accumulator and configuration object needed to audit the CSV files. Console output must be captured verbatim.

A reproduction is accepted only if:

(i) every input and source hash matches the manifest;
(ii) all exhaustive and integrity tests pass;
(iii) the fourteen observed values, exact means, and exact variances reproduce within the stated forward-error bounds;
(iv) the sign of every separation interval is negative;
(v) raw Monte Carlo summaries reproduce from the declared seeds, and an independent generator gives compatible confidence intervals;
(vi) the numerical tables are generated only from raw files;
(vii) every table cell in the manuscript is traceable to one raw column and row;
(viii) a fresh LaTeX build completes without undefined references, missing citations, or overfull material that obscures data.

The intended single command is

```
matlab -batch "run_all_reproductions"
```
(B.19)

from the archive root. The recorded full exact and Monte Carlo run required 2,133.7 seconds in the stated environment. The command and all programs in this appendix are specified for an independent rerun; they were not executed as part of the editorial production of Version 4.

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